








Modelling Financial Markets

Chapter 13: Machine Learning in Finance



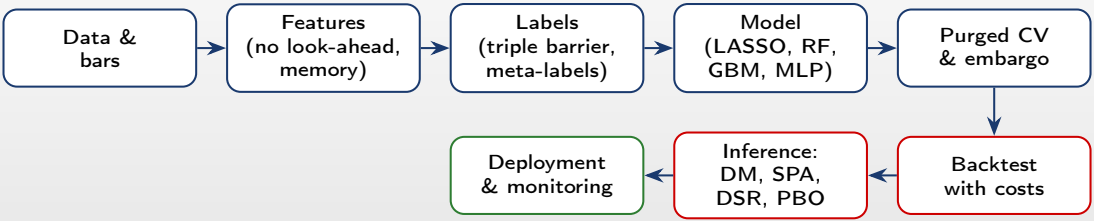
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What connects the steps of the analysis

- Part I: the prediction problem, a minimal toolbox, features and fractional differencing, labels, honest validation
- Part II: an S&P 500 case study with formal tests, selection and backtest overfitting, machine learning (ML) for inference, a research case study, artificial intelligence (AI) for scientific discovery (proofs and estimators: Appendix)
- Abbreviations in the diagram: LASSO (L_1-penalised regression), RF (random forest), GBM (gradient boosting machine), MLP (multilayer perceptron), CV (cross-validation)
- What you should be able to do after this chapter
  - build features and labels without look-ahead
  - validate a model with purging and embargo
  - test a classifier and a backtest against a benchmark: the Pesaran–Timmermann (PT) and Diebold–Mariano (DM) tests, the deflated Sharpe ratio (DSR), the superior predictive ability (SPA) test and the probability of backtest overfitting (PBO)
- Most failures come from the process, not the algorithm (López de Prado (2018)); two failures dominate
  - **look-ahead bias**: using information not available at decision time
  - **selection bias**: reporting the best of many trials

What We Will Explain: from Data to a Tested Claim



- Features use only the information available at the decision date
- Validation measures what still holds on observations the model has not seen
- A predicted direction justifies a trade only after uncertainty and costs are assessed

Why Machine Learning in Finance, and Why It Is Hard

The promise

- Flexible, non-linear functional forms; interactions found automatically
- Hundreds of candidate predictors (the "factor zoo", Chapter 3)
- Alternative data: text, images, order books, on-chain data

The constraints

- Very low **signal-to-noise ratio**; one serially dependent history that cannot be re-run
- **Non-stationarity**: regimes, breaks; markets react to their users
- Labels must be constructed; errors cost real money
- Massive risk of **overfitting** (fitting noise)

Adaptive Markets Hypothesis: Lo (2004)

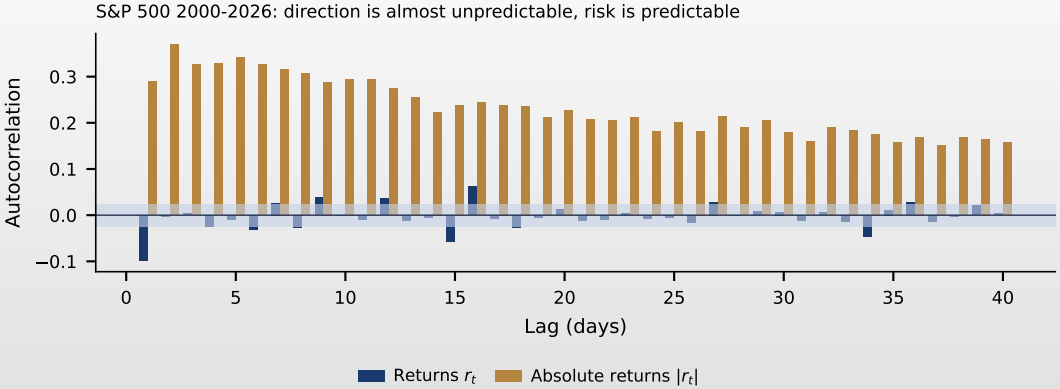
- Predictability appears, is arbitraged away and re-appears: a model that worked yesterday is a **hypothesis**, not a law (market efficiency: Chapter 2)
- The hard part is not fitting a model but **not fooling yourself**: correct labels, correct validation, honest backtests

What Is Predictable? Direction vs. Risk (1/2)

- The question: which part of a financial series is predictable at all?
- Autocorrelation at lag k, $\rho_k = \mathrm{Corr}(x_t, x_{t-k})$, for the daily log return $r_t = \log P_t - \log P_{t-1}$ and for $|r_t|$
- Next chart: lag k on the horizontal axis, $\hat{\rho}_k$ on the vertical axis; blue bars for r_t, brown bars for $|r_t|$
- The light-blue band, $\pm 1.96/\sqrt{T}$ (T: number of observations), is the 95% band under no autocorrelation
- Autocorrelations of returns r_t are tiny (inside the band); those of $|r_t|$ are large and persistent
- ML has the best chance where there is structure: **volatility, risk, liquidity, cross-sectional ranking**

MFM_ch13_model_comparison

What Is Predictable? Direction vs. Risk (2/2)



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Evidence, and the Economic Value of a Small R^2_{OOS} (1/2)

- Gu, Kelly & Xiu (2020): monthly returns of individual US stocks, 1957–2016; they compare linear models, penalised regressions, principal component regression (PCR), partial least squares (PLS), random forests, gradient boosting and neural nets
- Their fit measure, the out-of-sample (OOS) R^2: how much of the squared return a forecast removes, relative to forecasting zero

$$R^2_{OOS} = 1 - \frac{\sum_t (r_t - \hat{r}_t)^2}{\sum_t r_t^2}$$

  - r_t: realised return in month t of the test period; $\hat{r}_t$: forecast made with information up to $t-1$
  - the denominator is the error of the zero forecast $\hat{r}_t = 0$: $R^2_{OOS} > 0$ if the model beats it, $R^2_{OOS} < 0$ if it does worse
- Results (Table 1, monthly R^2_{OOS}): three-layer network 0.40%, gradient boosting 0.34%, random forest 0.33%, OLS with 3 predictors 0.16%, OLS with all 900+ predictors −3.46%
- Trees and nets win through **non-linear interactions**; momentum, liquidity and volatility dominate
- The benchmark matters: Welch & Goyal (2008) show that most published predictors lose to the **historical mean**; Gu, Kelly & Xiu (2020) and Chapter 14 use a zero forecast

Evidence, and the Economic Value of a Small R^2_{OOS} (2/2)

- Campbell & Thompson (2008): what a small R^2 is worth to an investor; returns have a predictable and an unpredictable part

$$r_{t+1} = \mu + x_t + \varepsilon_{t+1}, \qquad R^2 = \frac{\mathrm{Var}(x_t)}{\mathrm{Var}(r_{t+1})}$$

  - μ: mean return; x_t: predictable component, known at t, mean zero; ε_{t+1}: unpredictable shock
  - R^2: share of the return variance explained by the predictor
- An unconstrained mean-variance investor who knows x_t (no costs, no estimation error) reaches the Sharpe ratio SR_x

$$SR_x^2 = \frac{SR_{BH}^2 + R^2}{1 - R^2}$$

  - Sharpe ratio (SR): mean excess return divided by its standard deviation
  - SR_{BH}: SR of buy-and-hold, which ignores x_t; $R^2 = 0$ gives $SR_x = SR_{BH}$
- S&P 500, monthly log returns 2001–2026: $SR_{BH} = 0.128$ a month (0.44 annualised with $\sqrt{12}$); $R^2 = 0.5\%$ a month $\Rightarrow SR_x = 0.51$ annualised (+15%), $R^2 = 1\% \Rightarrow 0.57$ (about +27%)
- For a **single index** the effective sample is far smaller than for thousands of stocks: the situation of the S&P 500 case study

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Where the Campbell–Thompson Formula Comes From

- Same model: $r_{t+1} = \mu + x_t + \varepsilon_{t+1}$, $\mathbb{E}[x_t] = 0$, ε_{t+1} independent of x_t; write $\sigma_r^2 = \mathrm{Var}(r_{t+1})$
- Step 1, the risk the informed investor still bears: $\mathrm{Var}(r_{t+1}) = \mathrm{Var}(x_t) + \sigma_\varepsilon^2$ and $R^2 = \mathrm{Var}(x_t)/\sigma_r^2$, so

$$\sigma_\varepsilon^2 = (1 - R^2)\,\sigma_r^2$$

- Step 2, the investor's conditional Sharpe ratio at t: mean $\mu + x_t$, variance σ_ε^2, squared SR $(\mu + x_t)^2/\sigma_\varepsilon^2$
- Step 3, average over t: $\mathbb{E}[(\mu + x_t)^2] = \mu^2 + \mathrm{Var}(x_t) = \mu^2 + R^2\sigma_r^2$
- Step 4, divide and use $SR_{BH} = \mu/\sigma_r$:

$$SR_x^2 = \frac{\mu^2 + R^2\sigma_r^2}{(1 - R^2)\,\sigma_r^2} = \frac{SR_{BH}^2 + R^2}{1 - R^2}$$

- So SR_x^2 is the average squared *conditional* Sharpe ratio of an investor who re-optimises each month, in the units of the data: monthly here
- Numbers, $R^2 = 0.005$: $SR_x^2 = (0.128^2 + 0.005)/0.995 = 0.02149$, $SR_x = 0.1466$ a month; $\times\sqrt{12} = 0.51$ a year; $0.1466/0.128 = 1.15$

Loss, Penalty and Five Model Families (1/5)

- The question of this section: why, with a weak signal, does variance control matter more than flexibility?
- Supervised learning picks, within a model family, the function that fits the training data best after a penalty for complexity

$$\hat{f} = \arg\min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} L\big(y_i, f(\mathrm{x}_i)\big) + \lambda\, \Omega(f)$$

  - (y_i, x_i), $i = 1, \ldots, n$: target and feature vector of observation i; $\mathcal{F}$: the model family (e.g. all trees of depth 3)
  - L: loss, the squared error $(y - f)^2$ or, for a probability f of $y = 1$, the log-loss $-[y \log f + (1 - y) \log(1 - f)]$
  - $\Omega(f)$: complexity penalty with weight $\lambda \geq 0$; $\lambda = 0$: no penalty, large λ: a simpler model
- λ is chosen by K-fold cross-validation (CV) on folds $\mathcal{P}_1, \ldots, \mathcal{P}_K$:

$$CV(\lambda) = \frac{1}{K} \sum_{k=1}^{K} \frac{1}{n_k} \sum_{i \in \mathcal{P}_k} L\big(y_i, \hat{f}_\lambda^{(-k)}(\mathrm{x}_i)\big), \qquad \hat{\lambda} = \arg\min_{\lambda} CV(\lambda)$$

  - n_k: number of observations in fold k; $\hat{f}_\lambda^{(-k)}$: the model fitted on the other $K - 1$ folds

Loss, Penalty and Five Model Families (2/5)

- Why we penalise: the expected test error at a point x_0, for $y = f(x) + \varepsilon$ with $\mathbb{E}[\varepsilon] = 0$ and $\mathrm{Var}(\varepsilon) = \sigma^2$ (Hastie, Tibshirani & Friedman (2009), Ch. 7):

$$\mathbb{E}\big[(y_0 - \hat{f}(x_0))^2\big] = \underbrace{\sigma^2}_{\text{noise}} + \underbrace{\big(\mathbb{E}[\hat{f}(x_0)] - f(x_0)\big)^2}_{\text{bias}^2} + \underbrace{\mathrm{Var}\big(\hat{f}(x_0)\big)}_{\text{variance}}$$

  - $\mathbb{E}$: expectation over possible training samples and over the noise of the new observation $y_0 = f(x_0) + \varepsilon_0$
  - bias: the systematic error of a too-simple family
  - variance: the sensitivity of $\hat{f}$ to the training sample; it grows with model flexibility
  - σ^2: variance of the unpredictable part of y, which no model can reduce
- Derivation: add and subtract $\mathbb{E}[\hat{f}(x_0)]$ in $y_0 - \hat{f}(x_0)$; the cross terms have mean zero because ε_0 is independent of the training sample
- In finance σ^2 dominates: an R^2 below 1% means that over 99% of the error is noise, so penalties and tree averaging cut variance at the price of a small bias

Loss, Penalty and Five Model Families (3/5)

- Penalised regressions, with p standardised features and coefficients $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)$:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \; \frac{1}{n} \sum_{i=1}^{n} \left(y_i - \mathrm{x}_i^\top \boldsymbol{\beta}\right)^2 + \lambda \left[\alpha \|\boldsymbol{\beta}\|_1 + \tfrac{1-\alpha}{2} \|\boldsymbol{\beta}\|_2^2\right]$$

  - $\|\boldsymbol{\beta}\|_1 = \sum_j |\beta_j|$, $\|\boldsymbol{\beta}\|_2^2 = \sum_j \beta_j^2$; $\alpha \in [0, 1]$: share of the absolute-value penalty
- Ridge, $\alpha = 0$ (Hoerl & Kennard (1970)): $\hat{\boldsymbol{\beta}} = \left(\mathrm{X}^\top \mathrm{X} + \frac{n\lambda}{2} \mathrm{I}\right)^{-1} \mathrm{X}^\top \mathrm{y}$; shrinks all coefficients, none exactly to zero, and stabilises the estimate when predictors are correlated
- LASSO (least absolute shrinkage and selection operator), $\alpha = 1$ (Tibshirani (1996)): the penalty $|\beta_j|$ has a kink at zero, so some coefficients become exactly zero: LASSO selects features
- Elastic net, $0 < \alpha < 1$ (Zou & Hastie (2005)): keeps groups of correlated predictors together, where LASSO arbitrarily keeps one of them
- The chapter's linear benchmark is the penalised logit: $p = P(y = 1 \mid \mathrm{x}) = 1/\left(1 + e^{-\mathrm{x}^\top \boldsymbol{\beta}}\right)$, with the log-loss in place of the squared loss

Loss, Penalty and Five Model Families (4/5)

- Trees: at each node v choose the feature j and threshold s that reduce the Gini impurity $G = 1 - \sum_c \hat{\pi}_c^2$ the most

$$(j^*, s^*) = \arg\min_{j,\, s} \left[\frac{n_L}{n_v} G(L) + \frac{n_R}{n_v} G(R) \right], \qquad L = \{x_j \le s\},\ R = \{x_j > s\}$$

  - $\hat{\pi}_c$: share of class c in the node; n_v, n_L, n_R: observations in the node and in its two children
  - $G = 0$ for a pure node; with two classes $G = 2\hat{\pi}(1 - \hat{\pi})$, at most 0.5 at 50/50; a leaf predicts the share of class 1

- Random forest (RF, Breiman (2001)): $\hat{f}_{RF}(x) = \frac{1}{B} \sum_{b=1}^{B} T_b(x)$, each tree T_b grown on a bootstrap sample with a random subset of features at each split

$$\mathrm{Var}\big(\hat{f}_{RF}(x)\big) = \rho\, \sigma_T^2 + \frac{1 - \rho}{B}\, \sigma_T^2$$

  - σ_T^2: variance of one tree; ρ: correlation between two trees (Hastie, Tibshirani & Friedman (2009), Ch. 15)
  - random feature subsets lower ρ, the only term that more trees cannot remove
  - the out-of-bag error is **optimistic** when labels overlap

Q Random Forests

Loss, Penalty and Five Model Families (5/5)

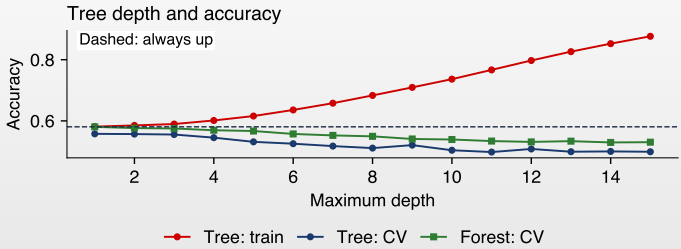
- Gradient boosting machine (GBM, Friedman (2001)): add, step by step, a small tree that corrects the errors of the current model

$$r_{im} = -\left.\frac{\partial L(y_i, F)}{\partial F}\right|_{F=F_{m-\mathbf{1}}(\mathbf{x}_i)}, \qquad F_m(\mathbf{x}) = F_{m-\mathbf{1}}(\mathbf{x}) + \eta\, h_m(\mathbf{x})$$

  - F_m: the model after m steps; h_m: the tree fitted to the pseudo-residuals r_{im}; for the log-loss on the logit score, $r_{im} = y_i - \hat{p}_{m-\mathbf{1}}(\mathbf{x}_i)$
  - $\eta \in (0, 1]$: learning rate; a small η learns slowly and overfits less

- Multilayer perceptron (MLP): hidden layers $\mathbf{h}^{(\ell)} = g\big(\mathbf{W}^{(\ell)}\mathbf{h}^{(\ell-\mathbf{1})} + \mathbf{b}^{(\ell)}\big)$, $\mathbf{h}^{(\mathbf{0})} = \mathbf{x}$, and the output $\hat{p} = 1/\big(1 + e^{-(\mathbf{w}^\top \mathbf{h}^{(D)} + b)}\big)$
  - $\mathbf{W}^{(\ell)}$, $\mathbf{b}^{(\ell)}$: weights and biases of layer ℓ; D: number of hidden layers; $g(z) = \max(0, z)$: the ReLU (rectified linear unit), element by element
  - the weights minimise the log-loss by gradient descent; early stopping ends training when the validation loss stops falling
  - richer architectures are in Chapter 14

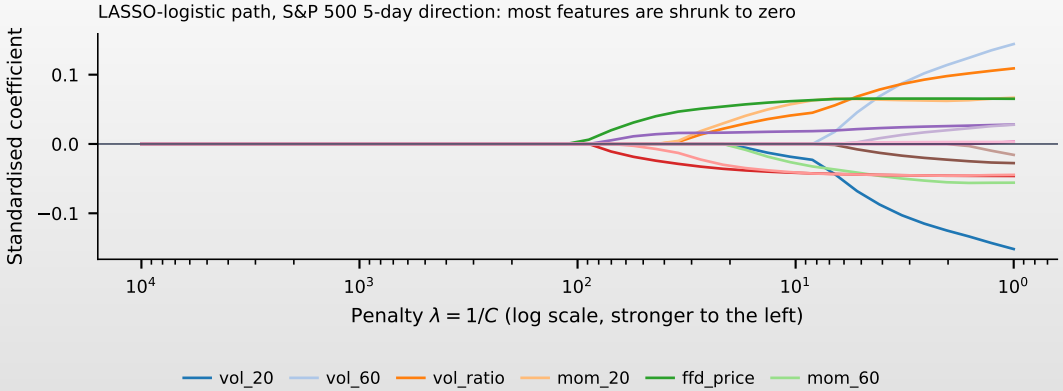
Bias–Variance in Practice: Tree Depth



- S&P 500, 5-day direction: training accuracy rises to $\approx 87\%$ with depth, while purged-CV accuracy **falls** towards 50%
  - purged CV drops from training the observations whose labels overlap the test set (section Honest Validation)
- Horizontal axis: maximum tree depth; vertical axis: accuracy; red training, blue CV of one tree, green CV of the forest
- Random forests degrade more slowly, but never beat the "always up" baseline (dashed line)

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LASSO Regularisation Path (1/2)



MFM_ch13_regularization

LASSO Regularisation Path (2/2)

- L_1-penalised logistic regression for the 5-day direction of the S&P 500, 14 standardised features
- Chart labels: vol_20, vol_60, vol_ratio are the 20- and 60-day volatilities and their ratio; mom_20, mom_60 the log returns over the last 20 and 60 days; ffd_price the FFD log price
- Horizontal axis: the penalty $\lambda = 1/C$, on a log scale, stronger to the left; vertical axis: the standardised coefficient
- Strong penalty (left of the chart): every coefficient is zero; the first feature to enter is the FFD (fixed-width window fractional differentiation) log price, a scaled price level (next section)
- Weak penalty: the volatility features get the largest coefficients
- All coefficients are small in absolute value: weak signal, consistent with near-efficient markets

MFM_ch13_regularization

Stationarity, Memory and the Fractional Difference (1/2)

- The question of this section: how do we turn a price into a stationary feature without erasing its memory?
- Log prices $X_t = \log P_t$ (P_t: closing price on day t) behave as $I(1)$; returns $\Delta \log P_t = \log P_t - \log P_{t-1}$ are $I(0)$ but **erase the memory** of the price level
- Integration order: a series X_t is $I(\delta)$ if $(1-B)^\delta X_t$ is $I(0)$
  - B: backshift operator, $BX_t = X_{t-1}$
  - $I(0)$: stationary with short memory; $I(1)$: a random walk type series, stationary after one difference
  - δ may be fractional: the series is stationary only for $\delta < \frac{1}{2}$, with long memory when $0 < \delta < \frac{1}{2}$ (Hosking (1981))
- Idea of fractional differencing: difference by less than 1, removing the unit root while keeping part of the memory

Stationarity, Memory and the Fractional Difference (2/2)

Definition 1 (Fractional difference of order d)

- $\tilde{X}_t = (1-B)^d X_t = \sum_{k=0}^{\infty} w_k X_{t-k}$, with $w_0 = 1$ and $w_k = -w_{k-1}\frac{d-k+1}{k}$

- d: differencing order chosen by the analyst; k: lag in days; w_k: weight of the value k days back, from the binomial expansion of $(1-B)^d$
- $d = 1$: $w = (1, -1, 0, \dots)$, ordinary returns; $0 < d < 1$: weights decay hyperbolically, so memory is kept
- An $I(1)$ log price differenced with d is $I(1-d)$: non-stationary for $d \leq \frac{1}{2}$, stationary with long memory for $\frac{1}{2} < d < 1$
- The infinite weights sum to $(1-1)^d = 0$: the level cancels only in the limit (asymptotics in the Appendix)
- López de Prado (2018) proposes the minimum d at which an ADF (Augmented Dickey–Fuller) unit-root test rejects (test equation and critical value on the ADF slide); this section shows why that rule fails

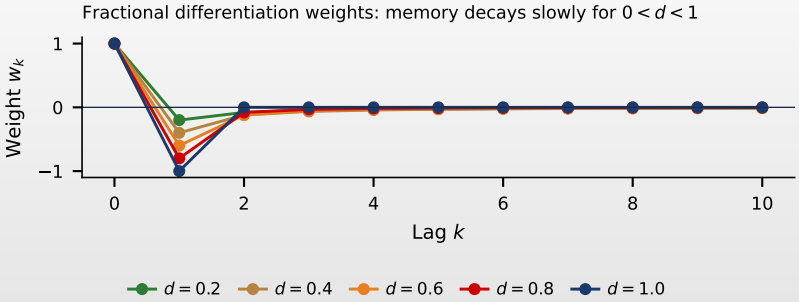
The Fixed-Width Window (FFD)

Definition 2 (Fixed-width-window fractional differentiation (FFD))

- Truncate the weights: keep $w_0, \ldots, w_K$ with $|w_k| \geq \tau$ (e.g. $\tau = 10^{-4}$), drop the rest (López de Prado (2018))
- $\tilde{X}_t = \sum_{k=0}^{K} w_k X_{t-k}$ for every $t > K$

- τ: smallest weight kept; K: the last lag kept, i.e. the window length; the first K days have too little history and are dropped
- Every $\tilde{X}_t$ uses the same number of lags; only past values enter $\Rightarrow$ no look-ahead
- The smaller d, the more lags are needed: $d = 0.25$, $\tau = 10^{-4} \Rightarrow K = 444$ lags
- The truncated weights no longer sum to zero: $\sum_{k=0}^{K} w_k = 0.178$ at $d = 0.25$ (0.040 at $d = 0.5$)

Fractional Differentiation Weights



- Horizontal axis: lag k; vertical axis: weight w_k, for five values of d
- For $d = 1$ only the first lag matters; for small d the weights of distant lags remain non-zero
- All weights after w_0 are negative: $\tilde{X}_t$ compares today's level with a weighted history

MFM_ch13_ffd

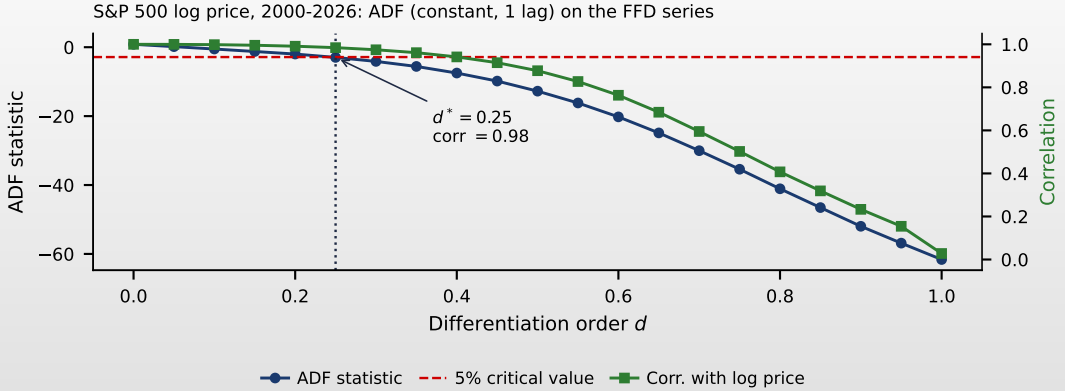
The ADF Statistic Along d: S&P 500 (1/3)

- Augmented Dickey–Fuller (ADF) unit-root test: the t-statistic of γ in

$$\Delta \tilde{X}_t = c + \gamma \tilde{X}_{t-1} + \sum_{i=1}^{p} \phi_i \Delta \tilde{X}_{t-i} + e_t$$

  - H_0: $\gamma = 0$, a unit root; p: number of lagged differences, which absorb the autocorrelation of e_t
  - with a constant, reject H_0 at 5% below ≈ -2.86
- The rule of López de Prado (2018): compute the FFD series on a grid of d between 0 and 1 and pick the smallest d at which the ADF rejects
- Next chart: d on the horizontal axis; the ADF statistic (blue) on the left axis; the correlation of the FFD series with the log price (green) on the right axis; red dashed line: the 5% critical value; vertical line: $d^* = 0.25$

The ADF Statistic Along d: S&P 500 (2/3)



MFM_ch13_ffd

The ADF Statistic Along d: S&P 500 (3/3)

- FFD log price, S&P 500 2000–2026, constant and 1 lag: ADF $= -2.96$, so the rule rejects from $d = 0.25$
- At $d = 0.25$, $\mathrm{Corr}(\tilde{X}_t, \log P_t) = 0.98$: the FFD series is almost the log price
- Same series, with 34 lags chosen by the Akaike information criterion (AIC): ADF $= -0.72$, no rejection
- The verdict depends on the number of lags, and a correlation of 0.98 with an $I(1)$ price is a warning, not a virtue

MFM_ch13_ffd

Truncated FFD of an $I(1)$ Price Is Still $I(1)$

- Decompose the FFD series:

$$\tilde{X}_t = \sum_{k=0}^{K} w_k X_{t-k} = \Big(\sum_{k=0}^{K} w_k\Big) X_t - \sum_{k=1}^{K} w_k \left(X_t - X_{t-k}\right)$$

  - first term: today's level times the sum of the truncated weights; second term: a weighted sum of the k-day changes $X_t - X_{t-k}$
- If X_t is $I(1)$, each $X_t - X_{t-k}$ is $I(0)$; the first term is $0.178\,X_t$ at $d = 0.25$: a **scaled random walk**
- The unit-root component has small variance next to the stationary part, so ADF with few lags confuses the two
  - Monte Carlo: pure random walk, $T = 6{,}274$ days, FFD with $d = 0.25$: ADF (1 lag) rejects in **97.8%** of 500 samples, at a nominal 5%
- A finite filter whose weights do not sum to zero keeps a level component, whatever the ADF p-value
- Consequence: the FFD log-price in the case study is a scaled price level, not a stationary feature; the next slides measure the memory directly

MFM_ch13_inference

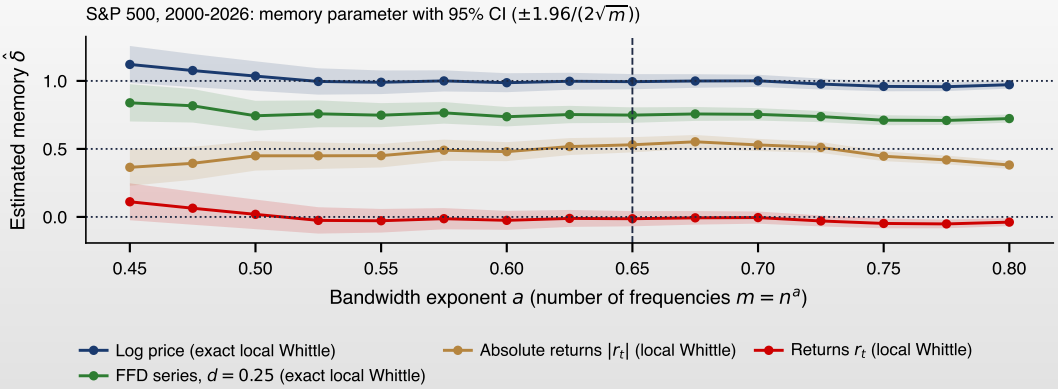
Memory Estimated Directly: Local Whittle (1/3)

- Idea: a long-memory series has most of its variance in slow cycles; the slope of its spectrum near frequency 0 reveals δ

$$f(\lambda) \sim G\,\lambda^{-2\delta} \quad \text{as } \lambda \to 0$$

  - $f(\lambda)$: spectral density, the variance contributed by cycles of frequency λ (period $2\pi/\lambda$ days)
  - $G > 0$: scale constant; $\sim$: the ratio of the two sides tends to 1
  - $\delta > 0$: f explodes near 0 (long memory); $\delta = 0$: flat (short memory); $\delta < 0$: f vanishes near 0
- Local Whittle (LW, Robinson (1995)): fits this law only on the first $m = n^a$ Fourier frequencies $\lambda_j = 2\pi j/n$, $j = 1, \ldots, m$
  - n: sample size; $a \in (0, 1)$: bandwidth exponent; a larger a uses more frequencies: less variance, more bias from short-run dynamics
- Exact local Whittle (ELW, Shimotsu & Phillips (2005)): also valid for non-stationary $\delta \geq \frac{1}{2}$; both estimators in the ▸ Appendix
- 95% confidence interval (CI): $\hat{\delta} \pm 1.96/(2\sqrt{m})$; the standard error $1/(2\sqrt{m})$ depends only on m

Memory Estimated Directly: Local Whittle (2/3)



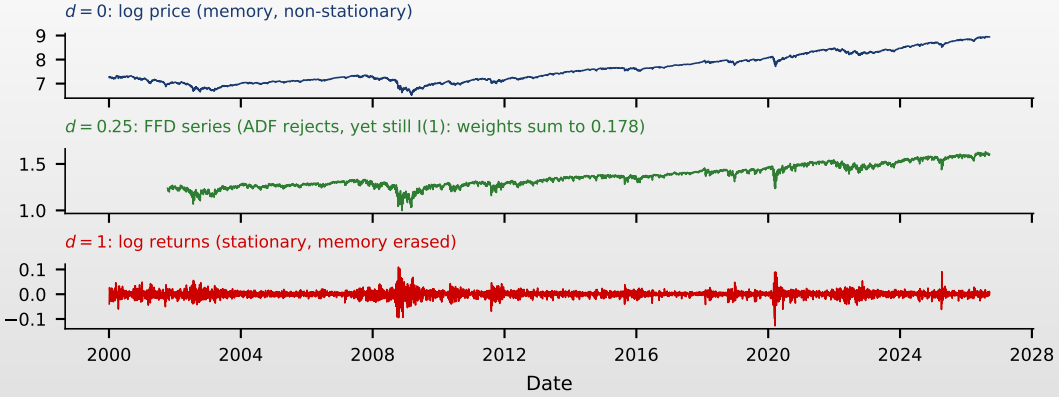
MFM_ch13_ffd

Memory Estimated Directly: Local Whittle (3/3)

- Chart: the exponent a (with $m = n^a$) on the horizontal axis, $\hat{\delta}$ on the vertical axis, with its 95% CI band; the vertical line marks $a = 0.65$, the value used below (S&P 500, 2000–2026)
- Log price (ELW): $\hat{\delta} = 0.99$, CI [0.94; 1.05]
- FFD series (ELW): $\hat{\delta} = 0.75$, CI [0.69; 0.81]
- Returns: $\hat{\delta} = -0.01$, CI [−0.07; 0.04]
- Absolute returns $|r_t|$: $\hat{\delta} = 0.53$, CI [0.47; 0.59]
- The FFD series stays near $1 - d$ over the frequency band used, although its integration order is still 1
- The long memory is in **volatility** (Chapters 5 and 9), so fractional filters make sense for $|r_t|$, not for the price

MFM_ch13_ffd

Prices, FFD Series and Returns (1/2)



MFM_ch13_ffd

Prices, FFD Series and Returns (2/2)

- The top panel shows the S&P 500 log price, an $I(1)$ series
- The middle panel shows the FFD series with $d = 0.25$: the price trend is still visible, because the series is $0.178 \log P_t$ plus a stationary part
- The bottom panel shows log returns, with no trend; only volatility clustering is visible

MFM_ch13_ffd

From Fixed-Horizon to Triple-Barrier Labels (1/2)

- The question of this section: which label reproduces a trader's real decision, with take-profit, stop-loss and a maximum holding period?
- **Fixed horizon**: the label is the sign of the h-day log return, with a dead zone around zero

$$r_{t,t+h} = \log P_{t+h} - \log P_t, \qquad y_t = \begin{cases} +1 & \text{if } r_{t,t+h} > \tau \\ -1 & \text{if } r_{t,t+h} < -\tau \\ 0 & \text{otherwise} \end{cases}$$

  - P_t: closing price on day t; h: horizon in days; $\tau \geq 0$: threshold, moves smaller than τ get the label 0
- Three problems
  - a constant τ ignores time-varying volatility
  - the path is ignored: a stop-loss could have been hit before $t + h$
  - the classes are unbalanced: S&P 500, $h = 10$, $\tau = 0$: 60.5% up vs. 39.5% down

From Fixed-Horizon to Triple-Barrier Labels (2/2)

Definition 3 (Triple-barrier label, López de Prado (2018))

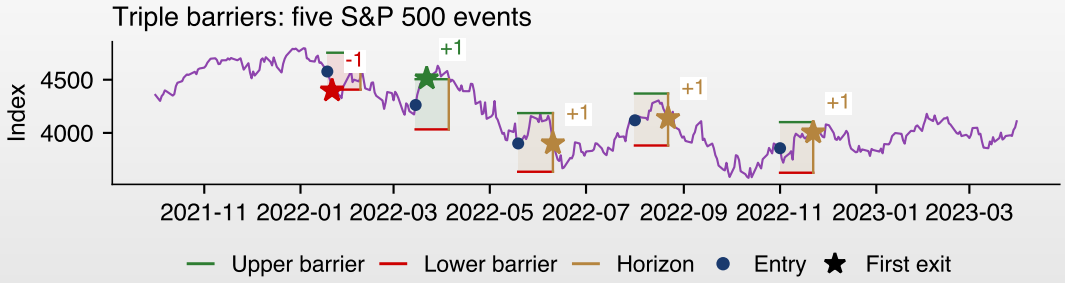
- At event time t_0, with daily volatility σ_{t_0} and horizon h: upper barrier $\log P_t - \log P_{t_0} \geq \pi_u \sigma_{t_0} \sqrt{h}$ (profit-taking), lower barrier $\leq -\pi_l \sigma_{t_0} \sqrt{h}$ (stop-loss), vertical barrier $t = t_0 + h$
- Label = **first barrier touched** at t_1: $+1$ upper, -1 lower; at the vertical barrier $\text{sign}(r_{t_0,t_1})$ (or 0)

- $\sigma_{t_0} \sqrt{h}$: the typical h-day move; $\pi_u, \pi_l > 0$: barrier widths in units of it
- Each label has a **span** $[t_0, t_1]$: the input of purging in validation
- σ_{t_0}: square root of the exponentially weighted moving average (EWMA) of squared deviations of daily log returns, span 50

$$\sigma_t^2 \approx \alpha \sum_{k \geq 0} (1-\alpha)^k \left(r_{t-k} - \bar{r}_t\right)^2, \qquad \alpha = \frac{2}{\text{span} + 1} = \frac{2}{51} \approx 0.039$$

  - $\bar{r}_t$: the exponentially weighted mean return; span 50: an effective memory of about two trading months
- Why volatility-scaled barriers: a fixed threshold would be hit almost daily in 2008 or March 2020 and rarely in calm years

Triple Barrier: S&P 500 in 2022 (1/2)

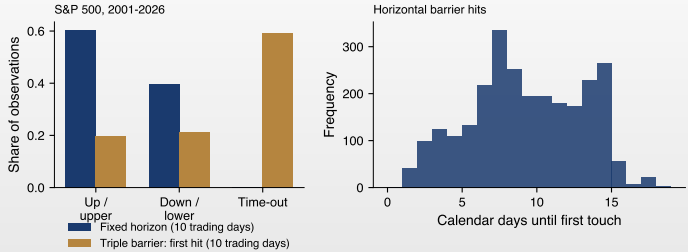


Triple Barrier: S&P 500 in 2022 (2/2)

- Five events during the 2022 bear market, $\pi_u = \pi_l = 1$, $h = 15$ days; the colour of each star is the barrier touched first
- Chart: the index level; green lines: upper barrier, red: lower barrier, brown: horizon; dots: entry, stars: first barrier touched
- January: stop-loss hit (-1); March: profit-taking ($+1$); the other three end at the vertical barrier (amber stars) above their entry level, so the sign rule labels them $+1$
- A label $+1$ therefore does not mean that the profit barrier was touched

MFM_ch13_triple_barrier

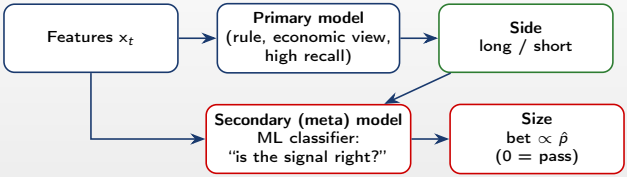
Label Distribution: Fixed Horizon vs. Triple Barrier



- S&P 500, 2001–2026, $h = 10$, barriers at $\pm 1\,\sigma\sqrt{h}$: 19.6% up, 21.3% down, 59.0% time-out, so balanced directional labels (fixed horizon: 60.5% up)
- The chart counts **exit types**: time-out = exit at the vertical barrier; under the sign rule these 59.0% still get ± 1 from the sign of their return, under the "0" convention they form a third class
- Right panel: histogram of calendar days until the first horizontal barrier is touched, typically after 1–2 weeks

MFM_ch13_triple_barrier

Meta-Labelling: Deciding Whether to Act (1/2)



- Meta-label $y_t^{meta} = 1$ if the primary signal was profitable (barrier touched in its direction), 0 otherwise
- The secondary model estimates $\hat{p} = P(y_t^{meta} = 1 \mid x_t)$, the probability that the signal of day t is right given the features x_t
- ML learns **when to trust** an interpretable primary rule, not the direction itself

Meta-Labelling: Deciding Whether to Act (2/2)

- A filter decides which signals to act on; three scores describe it

$$\text{precision} = \frac{TP}{TP + FP}, \qquad \text{recall} = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

  - TP: profitable signals taken; FP: unprofitable signals taken; FN: profitable signals skipped
  - acting on every signal gives recall 1; F_1: their harmonic mean, between 0 and 1
- Example (Seminar 13, exercise B11): the 20- and 50-day moving-average crossover (MA20/MA50) on Bitcoin, 1,660 test trades
  - with the meta filter, precision rises from 0.475 to 0.517, recall falls from 1 to 0.541, and F_1 falls from 0.644 to 0.529
  - the mean gross log return per trade, before costs, moves from -0.27% to $+0.19\%$
- Meta-labelling raises precision at the cost of recall; with costs per trade, precision and the return per trade matter, not F_1

Overlapping Labels and Sample Uniqueness (1/2)

- Label i spans $[t_{0,i}, t_{1,i}]$; consecutive labels share returns $\Rightarrow$ samples are **not i.i.d.**
- Concurrency: the number of labels whose span contains day t

$$c_t = \sum_i 1\{t_{0,i} \leq t \leq t_{1,i}\}$$

- Average uniqueness of label i: the average share of each day of its span that belongs to it alone

$$\bar{u}_i = \frac{1}{t_{1,i} - t_{0,i} + 1} \sum_{t=t_{0,i}}^{t_{1,i}} \frac{1}{c_t}$$

  - $\bar{u}_i = 1$: no shared return; $\bar{u}_i = 1/5$: the harmonic mean of c_t over the span of label i is five

Overlapping Labels and Sample Uniqueness (2/2)

Consequences

- Effective sample size $\approx \sum_i \bar{u}_i \ll n$, with n the number of labels (an overlap heuristic, not a variance-based measure)
- Bagging draws near-duplicates; CV scores too optimistic

Remedies

- Weight samples by $\bar{u}_i$ (and by $|r|$)
- Sequential bootstrap; in RF, the fraction of observations drawn for each tree $\approx \bar{u}$, the mean uniqueness

Leakage: What Crosses the Train/Test Boundary

- The question of this section: why does ordinary K-fold overstate a model's skill on financial series?

Definition 4 (Leakage)

- The training set contains information that is also present in the test set, so the test score no longer measures performance on genuinely new data

- Serially correlated features ($x_t \approx x_{t+1}$) and overlapping labels ($y_t \approx y_{t+1}$): a shuffled K-fold puts **near-identical** days on both sides
- A flexible model (RF, GBM) memorises the neighbour $\Rightarrow$ excellent "out-of-sample" score; tuning on a leaky CV selects the **most overfitted** model
- Other sources: transformations fitted on the full sample (scaling, principal component analysis, the choice of d for FFD), survivorship bias, data revisions, wrong timestamps

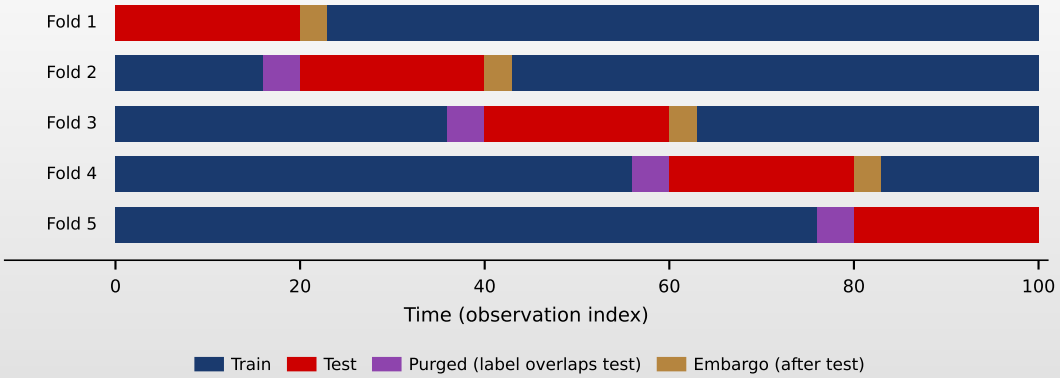
Purged K-Fold with Embargo (1/2)

- Folds are **contiguous** blocks in time (López de Prado (2018)); test fold k holds the observations $t_k^a, \ldots, t_k^b$, whose labels end at the latest at $T_k = \max_{i \in \text{test}_k} t_{1,i}$
- The training set keeps only the observations that do not touch the test:

$$\mathcal{A}_k = \left\{ i \notin \text{test}_k : \; [t_{0,i}, t_{1,i}] \cap [t_k^a, T_k] = \emptyset \;\; \text{and} \;\; t_{0,i} \notin (t_k^b, \, t_k^b + e] \right\}$$

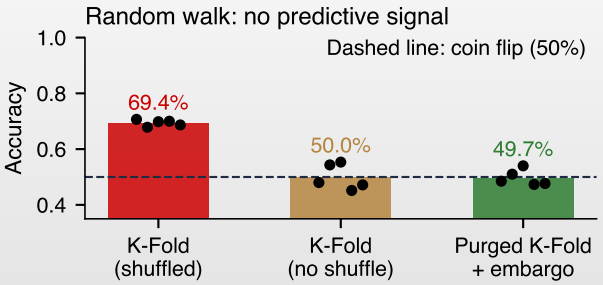
  - the first condition is **purging**: it removes labels that overlap the span of the test labels
  - the second is the **embargo** of e observations after the test block (e.g. 1%), needed because features look back in time
- The purged CV score is the mean of the K scores of models fitted on $\mathcal{A}_k$ and evaluated on fold k
- With labels of length h, a boundary removes h observations before and $\max(h, e)$ after the test block: the embargo adds anything only if $e > h$ (a 10-observation example: Appendix)
- Next chart: five folds in rows, observation index on the horizontal axis; blue train, red test, purple purged, brown embargo

Purged K-Fold with Embargo (2/2)



MFM_ch13_purged_cv

Leakage Experiment: Nothing to Predict



- Price = pure random walk
  - label = sign of the 20-day forward return
  - 5 features = 20-day rolling means of independent noise
- True accuracy is 50% by construction; random forest, 5 folds
- **Shuffled K-fold: 69.4%**; contiguous K-fold 50.0%; purged K-fold + embargo 49.7%
- Shuffling fails because labels overlap *and* features persist
  - here contiguous folds already remove most of the leak: only the fold boundaries share returns, and purging cleans them
  - a result that looks too good is usually leakage

MFM_ch13_purged_cv

Model Selection versus Deployment

- Ordinary K-fold is valid for a purely autoregressive model with serially uncorrelated errors (Bergmeir, Hyndman & Koo (2018)); when it fails: Appendix
- **Model selection**: purged K-fold, or combinatorial purged CV (CPCV)
  - CPCV tests on many combinations of groups and returns a **distribution** of Sharpe ratios instead of one number
- **Deployment**: walk-forward (WF)
  - train on $[0, T_k]$, test on $(T_k, T_{k+1}]$, roll forward (T_k: end of the k-th training window)
  - realistic, but a single path with high variance: it serves the final evaluation, not the choice (WF and CPCV: Appendix)

Case Study: Can the 5-Day Direction of the S&P 500 Be Predicted?

- S&P 500 index and VIX (the CBOE volatility index, the 30-day implied volatility of S&P 500 options), daily closes of the course data, 2001–2026: $n = 6{,}269$ observations
- Why the S&P 500: the most liquid equity index, with a long VIX history; if its direction were predictable, arbitrage would be cheap
- Target: $y_t = 1\{\log P_{t+5} - \log P_t > 0\}$, equal to 1 if the index is higher 5 days later: the fixed-horizon label with $h = 5$, $\tau = 0$
- Why 5 days: a weekly trading horizon, with about $6{,}269/5 \approx 1{,}250$ non-overlapping labels
- The label span $[t, t + 5]$ is used for purging
- Validation: purged 5-fold CV, 1% embargo, the same folds for every model; each fold covers about 5 years, hence at least one crisis episode

Case Study: Metrics, against the "Always Up" Baseline

- Accuracy: share of days whose direction is predicted correctly
- Receiver operating characteristic (ROC) area under the curve (AUC): the probability that a randomly chosen up day gets a higher predicted probability than a randomly chosen down day
- AUC $= 0.5$: no skill; AUC $= 1$: perfect ranking
- Example: up days with probabilities 0.7 and 0.4, down days with 0.5 and 0.2; 3 of the 4 pairs are ranked correctly, so AUC $= 0.75$
- Why both: accuracy depends on the 0.5 threshold, AUC compares only the ranking of the probabilities
- **Fold mean**: the metric computed in each of the 5 test folds, then averaged
- **Pooled**: the metric computed once on all out-of-fold predictions together; used by the formal tests

Case Study: 14 Features and Four Models

Features, information up to day t

- **Momentum**: $\log P_t - \log P_{t-h}$, the log return over the last h days, $h \in \{1, 5, 20, 60, 120\}$
- **Realised volatility**: annualised standard deviation (SD) of daily log returns over 20 and 60 days, and their ratio
- **Distance to moving average**: $\log P_t$ minus the mean of $\log P$ over 50 or 200 days
- **RSI(14)**: $100 - 100/(1 + RS)$, RS = smoothed average gain over average loss, 14 days; between 0 and 100, high after rises
- **VIX**: log level and 5-day change
- **FFD** log price, $d = 0.25$, chosen on the full sample (ablation after the model comparison)

Models, deliberately conservative

- **Logit**: L2-penalised (Ridge), standardised features
- **RF**: 400 trees, minimum leaf 100
- **GBM**: histogram-based, depth 3, learning rate 0.03, 200 trees
- **MLP**: hidden layers of 32 and 16 units, early stopping
- Large leaves and shallow trees limit variance

The Model That Learned "Always Up" (1/2)

	Pred. up	**Pred. down**
Actual up	TP = 3,572	FN = 68
Actual down	FP = 2,591	TN = 38

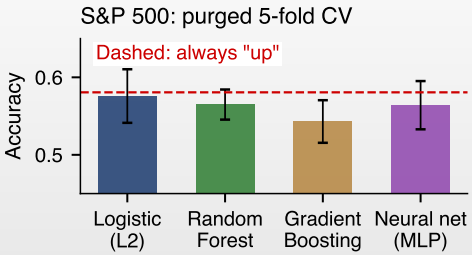
- L2-penalised logit, pooled out-of-fold predictions of purged 5-fold CV, $n = 6{,}269$
- TP, FN, FP, TN: true positives, false negatives, false positives, true negatives

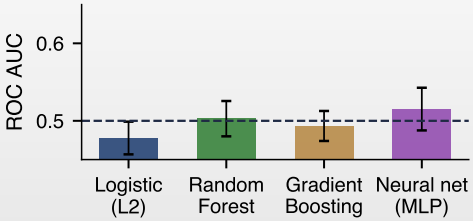
- Accuracy $= (TP + TN)/n = 57.6\%$
- Precision $= TP/(TP + FP) = 0.580$
- Recall $= TP/(TP + FN) = 0.981$
- $F_1 = 0.729$ (formula on the meta-labelling slide)
- The model predicts "up" on 98.3% of days: it has essentially learned "always up"

The Model That Learned "Always Up" (2/2)

- **Majority baseline** from the same table: actual up days $(3{,}572 + 68)/6{,}269 = 3{,}640/6{,}269 = 58.1\%$
  - the accuracy of predicting "up" every day, with no model at all
- A 57.6%-accurate model is *worse* than doing nothing
- The cause is class imbalance: with weak features the predicted probabilities stay near the base rate 0.58, above the 0.5 threshold, so almost every day is classified "up"
  - the case study keeps the fixed-horizon label on purpose, so that the baseline is transparent
  - triple-barrier labels change the class balance the classifier sees (Seminar 13, exercises B1 and B8)
- Other baselines: yesterday's direction, a moving-average rule, buy-and-hold for trading
- A model has value only if it beats the baseline **out of sample**, after costs, by more than sampling error

Model Comparison: Fold Means





Fold mean	Logit	RF	GBM	MLP
Accuracy	57.6%	56.5%	54.3%	56.4%
AUC	0.478	0.503	0.493	0.515

- Bars: mean over the 5 purged folds; whiskers: ± 1 SD across folds
- Baseline "always up": **58.1%**; no model beats it

MFM_ch13_model_comparison

Transformations on Training Data Only: the FFD Ablation

- The "minimum d" was chosen on 2000–2026 and then used inside purged CV: the test folds helped choose a feature
  - the same ADF rule on data up to 2009 gives $d = 0.15$; on the full sample $d = 0.25$
- Case study without the FFD log-price (purged 5-fold, mean over folds)
  - accuracy: logit 57.5%, RF 56.4%, GBM 55.6%, MLP 55.5%, all below 58.1%
  - AUC: logit 0.483, RF 0.522, GBM 0.514, MLP 0.514
- The negative conclusion does not depend on this feature
- Correct protocol
  - estimate d (local Whittle, with its CI) inside each training fold only
  - prefer features with an economic meaning: distances to moving averages, fractional filters of $|r_t|$

MFM_ch13_inference

Overlapping Labels and the Newey–West Long-Run Variance

- The tests below average a daily series z_t (a hit, a loss differential d_t, a regression score); with 5-day labels, neighbouring z_t share returns, so $\mathrm{Var}(\bar{z}) \neq \mathrm{Var}(z_t)/n$
- Newey–West (Newey & West (1987)) heteroskedasticity and autocorrelation consistent (HAC) long-run variance, with Bartlett weights:

$$\hat{\Omega}_{NW} = \hat{\Gamma}_{\mathbf{0}} + 2\sum_{j=\mathbf{1}}^{q}\Big(1 - \frac{j}{q+1}\Big)\hat{\Gamma}_j, \qquad \hat{\Gamma}_j = \frac{1}{n}\sum_{t=j+\mathbf{1}}^{n}(z_t - \bar{z})(z_{t-j} - \bar{z})$$

  - $\hat{\Gamma}_j$: sample autocovariance of z_t at lag j; q: number of lags ("lag 4" in the tables); $\widehat{\mathrm{SE}}(\bar{z}) = \sqrt{\hat{\Omega}_{NW}/n}$
  - lag rule: h-day labels share returns up to $h-1$ days apart, so $q \geq h-1 = 4$ is the minimum here
- Toy example: z_t = sum of the next 5 i.i.d. daily returns of variance $\sigma^{\mathbf{2}}$; $\Gamma_j = (5-j)\sigma^{\mathbf{2}}$ for $j \leq 4$, 0 beyond
  - i.i.d. formula: $\Gamma_{\mathbf{0}} = 5\sigma^{\mathbf{2}}$; true long-run variance $\Gamma_{\mathbf{0}} + 2(4+3+2+1)\sigma^{\mathbf{2}} = 25\sigma^{\mathbf{2}}$: an i.i.d. t is $\sqrt{5} \approx 2.2$ times too large
  - Bartlett with $q = 4$: $5 + 2(0.8 \cdot 4 + 0.6 \cdot 3 + 0.4 \cdot 2 + 0.2 \cdot 1) = 17\sigma^{\mathbf{2}}$; the weights damp far lags, so $q = h-1$ is a floor, not a generous choice

Are the Differences Significant? Pesaran–Timmermann and AUC Intervals (1/2)

- **Pesaran–Timmermann** (PT, Pesaran & Timmermann (1992)): are predicted and realised directions related?

$$PT = \frac{\hat{P} - \hat{P}^*}{\widehat{\mathrm{SE}}} \xrightarrow{d} N(0,1), \qquad \hat{P}^* = \hat{p}_y \hat{p}_x + (1 - \hat{p}_y)(1 - \hat{p}_x)$$

  - $\hat{P}$: share of days with the correct sign; $\hat{p}_y$, $\hat{p}_x$: shares of realised and of predicted "up"; $\hat{P}^*$: share correct expected if predictions were independent of outcomes
  - H_0: independence; $\widehat{\mathrm{SE}}$: standard error of $\hat{P} - \hat{P}^*$ under H_0 (Appendix)
  - $PT > 1.645$: skill at 5% (one-sided); $PT < 0$: fewer hits than chance
- PT assumes independent days; with overlapping labels we also use the slope t in $y_t = a + b\,x_t + u_t$, Newey–West errors with $q = 4$ (Newey–West slide): x_t the predicted and y_t the realised direction, $b > 0$ means skill
- **AUC**: the DeLong standard error (SE) assumes independent days (Appendix)
- Hence a circular block bootstrap on blocks of $L = 20$ consecutive days (Künsch (1989)), four times the label length, which keeps the serial dependence

Are the Differences Significant? Pesaran–Timmermann and AUC Intervals (2/2)

Pooled out-of-fold	Logit	RF	GBM	MLP
$\hat{P}$ / $\hat{P}^*$	0.576 / 0.578	0.565 / 0.570	0.543 / 0.557	0.564 / 0.565
PT / HAC t (lag 4)	-1.28 / -1.00	-1.61 / -1.16	-3.07 / -2.31	-0.36 / -0.26
AUC	0.479	0.495	0.490	0.512
block-bootstrap 95% CI	[0.455; 0.504]	[0.472; 0.520]	[0.468; 0.514]	[0.489; 0.537]

- No significant positive directional skill: every block-bootstrap AUC interval contains 0.5
- GBM is significantly *worse* at HAC lag 4 ($t = -2.31$), but not after a Bonferroni correction for the four models (threshold $|t| > 2.50$)
- Logit, for contrast: the DeLong 95% CI, which treats days as independent, is [0.465; 0.494] and excludes 0.5
- Consistent with weak-form efficiency (Chapter 2)
- Look for signal in cross-sections (Gu, Kelly & Xiu (2020)), in risk targets such as volatility (Chapters 5 and 9), in crypto delisting (Będowska-Sójka, Wójcik & Pele (2026)) and in text (Chapter 15)

MFM_ch13_inference

Comparing Forecasts: Diebold–Mariano, Clark–West, Giacomini–White (1/2)

- Two probability forecasts of $y_t \in \{0, 1\}$: model 1 gives $\hat{p}_{1,t}$, the benchmark (model 0, e.g. a constant) gives $\hat{p}_{0,t}$
- Loss of a forecast $\hat{p}_t$: the log-loss of the toolbox slide, with $f = \hat{p}_t$, or the Brier loss $(y_t - \hat{p}_t)^2$
- Loss differential $d_t = L(y_t, \hat{p}_{1,t}) - L(y_t, \hat{p}_{0,t})$; $d_t > 0$: model 1 worse on day t
- **Diebold–Mariano** (DM, Diebold & Mariano (1995)): is the mean loss differential zero?

$$DM = \frac{\bar{d}}{\sqrt{\hat{\Omega}_{NW}/n}} \xrightarrow{d} N(0,1) \quad \text{under } H_0 : \mathbb{E}[d_t] = 0$$

  - $\bar{d}$: mean of d_t over the n test days; $\hat{\Omega}_{NW}$: Newey–West long-run variance of d_t (Newey–West formula, with $z_t = d_t$)
  - $DM > 1.96$: model 1 significantly worse at 5%; $DM < -1.96$: significantly better

Comparing Forecasts: Diebold–Mariano, Clark–West, Giacomini–White (2/2)

- **Nested models** (the logit with 14 features nests the constant, $\beta = 0$): under H_0 the large model estimates zeros with noise, $\mathbb{E}[d_t] > 0$, and DM rarely favours it
- **Clark–West** (CW, Clark & West (2007)): one-sided HAC t-test of $\mathbb{E}[f_t] = 0$ against $\mathbb{E}[f_t] > 0$

$$f_t = (y_t - \hat{p}_{0,t})^2 - \left[(y_t - \hat{p}_{1,t})^2 - (\hat{p}_{0,t} - \hat{p}_{1,t})^2\right]$$

  - the Brier-loss gain of model 1, plus the term $(\hat{p}_{0,t} - \hat{p}_{1,t})^2$ that removes the noise cost of estimating zero coefficients; a large positive t: the features add information
  - derived for nested linear models estimated by ordinary least squares (OLS); for penalised-logit probabilities an approximation, not a theorem
- **Giacomini–White** (GW, Giacomini & White (2006)): tests the forecasting *method* as deployed, with a bounded estimation window
- By contrast, CW asks whether the population model adds information (the GW statistic in the Appendix)

Feature Importance: MDI and MDA

- The question: which features does the model rely on, and which of them help prediction on new data?
- Mean decrease impurity (MDI), trees only, computed in sample:

$$\mathrm{MDI}_j = \frac{1}{B}\sum_{b=1}^{B} \sum_{v \in T_b:\ j(v)=j} \frac{n_v}{n}\left[G_v - \frac{n_L}{n_v} G_L - \frac{n_R}{n_v} G_R\right]$$

  - the sum of the Gini reductions of the nodes v that split on feature j, weighted by the share n_v/n of observations reaching the node
  - favours continuous features with many possible thresholds
- Mean decrease accuracy (MDA), any model, on test data:

$$\mathrm{MDA}_j = S\big(\hat{f};\, \mathcal{D}_{test}\big) - \frac{1}{R}\sum_{r=1}^{R} S\big(\hat{f};\, \mathcal{D}_{test}^{\pi_r(j)}\big)$$

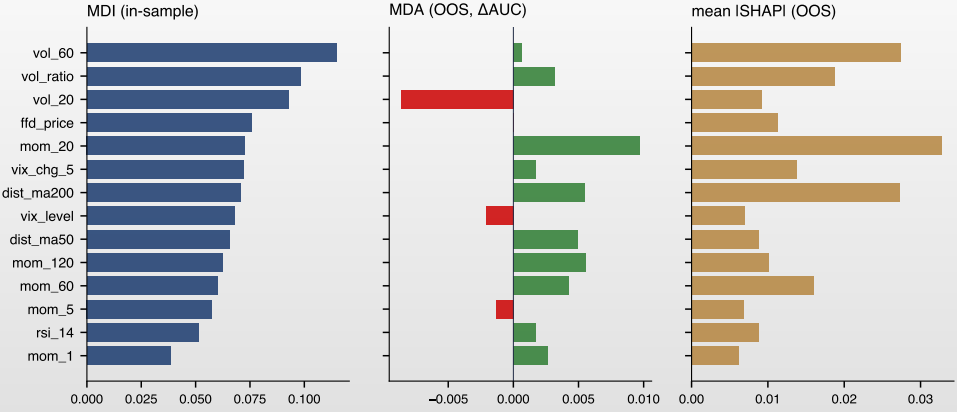
  - S: the score (here AUC); $\mathcal{D}_{test}^{\pi_r(j)}$: the test data with column j randomly permuted, repetition r
  - $\mathrm{MDA}_j < 0$: permuting raised the score, so the feature does not help; the value may also be sampling noise

Feature Importance: SHAP Values

- SHAP (Lundberg & Lee (2017)): the Shapley value $\phi_j(\mathrm{x})$ is the average marginal contribution of feature j to the prediction at x, averaged over all coalitions of other features (coalition formula: Appendix)
- Additive: $\hat{f}(\mathrm{x}) = \phi_0 + \sum_j \phi_j(\mathrm{x})$, ϕ_0 the average prediction
- Global importance of feature j: $\frac{1}{n}\sum_i |\phi_j(\mathrm{x}_i)|$, computed on test data
- MDI answers "what did the model use?"; MDA and SHAP answer "what helps on new data?"
- Correlated features share and dilute importance (substitution): cluster them first (López de Prado (2020))

Shapley Values

Feature Importance in the Case Study (1/2)



MFM_ch13_feature_importance

Feature Importance in the Case Study (2/2)

- Fit on the first 70%, evaluated on the last 30% after a purge; **RF** gives MDI and MDA, **GBM** gives SHAP, so the differences are not purely methodological
- Panels: MDI (in sample), MDA (out-of-sample AUC drop; red bars are negative) and mean $|\phi_j|$ (out of sample)
- Labels beyond the LASSO path: mom_h, h-day momentum; dist_ma50, dist_ma200, the distance to the moving average; rsi_14, the RSI(14); vix_level, vix_chg_5, the VIX level and its 5-day change
- MDI ranks volatility first; MDA and SHAP point to 20-day momentum and the distance from the 200-day moving average
- 20-day volatility: high MDI, **negative** MDA: a sign of overfitting, not a proof
- All importances are tiny ($\Delta\text{AUC} < 0.01$); how to read them: ▸ Appendix
- Correlated features: permutation creates off-manifold points, so MDA and SHAP **extrapolate** (Hooker, Mentch & Zhou (2021))

MFM_ch13_feature_importance

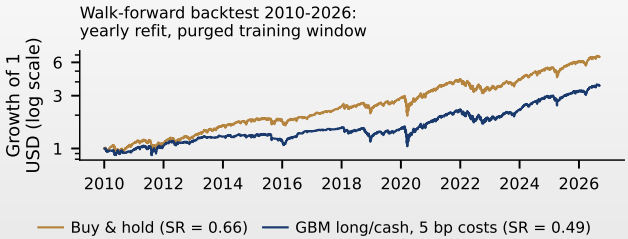
Walk-Forward Backtest: Strategy and Design

- The question: would a trader following the model have earned more, after costs, than by simply holding the index?
- Position $w_t \in \{0, 1\}$: long the S&P 500 if $\hat{p}_t > 0.5$, otherwise cash, decided with information up to t ($\hat{p}_t$: the model's probability of "up")
- Strategy return: the index return when invested, minus a cost each time the position changes

$$r^{s}_{t+\mathbf{1}} = w_t \, r_{t+\mathbf{1}} - c \, |w_t - w_{t-\mathbf{1}}|$$

  - $r_{t+\mathbf{1}}$: daily log return of the index; turnover $|w_t - w_{t-\mathbf{1}}|$: 1 on the days the position changes
  - $c = 5$ basis points (bp), i.e. 0.05% per unit traded: a prudent cost for an exchange-traded fund on the S&P 500
  - $r - c$ approximates $\log\left((1-c)\, e^{r}\right) = r + \log(1-c)$, the log return after paying the cost
- Exposure $\bar{w} = \frac{\mathbf{1}}{T} \sum_t w_t$: share of the T days spent invested
- Cash earns 0; both Sharpe ratios use daily log returns with risk-free rate 0, annualised with $\sqrt{252}$, so interest on cash days is ignored
- Model: GBM refitted every January, 2010–2026, on all past data, purging labels that overlap the test year
- Benchmark: buy-and-hold ($w_t = 1$: exposure 100%, no turnover)

Walk-Forward Backtest: Results



- 2010–2026: strategy Sharpe **0.49** vs. buy-and-hold **0.66** (annualised); average exposure 77%
- The model stayed mostly long; its exits from the market (23% of the days) did not improve the Sharpe ratio
- Chart: growth of one dollar invested, log scale; brown buy-and-hold, blue the GBM strategy
- The Sharpe-ratio difference, -0.17, has $p = 0.073$ (Ledoit–Wolf HAC test, detailed in the selection section)
- There is neither evidence of skill nor significant evidence that the strategy is worse
- With $d = 0.15$, chosen by the same rule on data up to 2009, SR $= 0.43$; without the FFD log-price 0.51

MFM_ch13_walk_forward

Walk-Forward: Do the Forecasts Beat a Constant?

- Walk-forward 2010–2026, $n = 4{,}198$ days, 60.2% "up"; benchmark: the constant forecast = training share of "up"; DM, Clark–West and GW use Newey–West with 4 lags
- Logit (nests the constant): log-loss 0.6822 vs. 0.6757
  - DM $t = 2.39$ ($p = 0.017$): the estimated logit is significantly *worse*
  - Clark–West $t = -0.21$ (one-sided $p = 0.58$): no evidence that the features carry information
  - Giacomini–White $= 5.77$ (χ^2_2, $p = 0.056$): no significant difference at 5%; approximate, since the window expands (rolling-window result in the Appendix)
- GBM: log-loss 0.6971; DM $t = 5.09$, GW $= 25.8$ ($p < 0.001$): worse than a constant; Pesaran–Timmermann -0.74
- Conclusion: no model beats "always up" or a constant forecast

MFM_ch13_inference

Selection Bias and the False Strategy Theorem

- The question of this section: how large is the best Sharpe ratio among N trials when none has skill?
- A researcher tries N strategies and reports the best; even if all have zero true Sharpe ratio, $\max_n \widehat{SR}_n$ grows with N: **selection bias**
- Harvey, Liu & Zhu (2016): given hundreds of published factors, a new factor needs $t > 3$, not 2

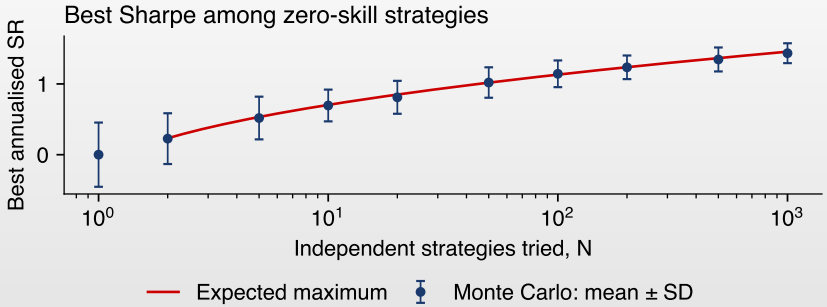
Theorem 1 (False Strategy Theorem, Bailey & López de Prado (2014))

- If $\widehat{SR}_n$, $n = 1, \ldots, N$, are independent and approximately $N\big(0, \mathrm{Var}[\widehat{SR}]\big)$ (no skill), the **expected maximum Sharpe ratio** is

$$SR_{\mathbf{0}} \equiv \mathbb{E}\Big[\max_n \widehat{SR}_n\Big] \approx \sqrt{\mathrm{Var}[\widehat{SR}]}\left[(1-\gamma_E)\,\Phi^{-\mathbf{1}}\Big(1-\tfrac{\mathbf{1}}{N}\Big) + \gamma_E\,\Phi^{-\mathbf{1}}\Big(1-\tfrac{\mathbf{1}}{Ne}\Big)\right]$$

- $\widehat{SR}_n$: estimated Sharpe ratio of strategy n; $\mathrm{Var}[\widehat{SR}]$: sampling variance of one estimated SR, $\approx 1/T$ per period for i.i.d. returns, T the number of return periods
- Φ: standard Normal distribution function; $\Phi^{-\mathbf{1}}$: its quantile function; $\gamma_E \approx 0.5772$: Euler–Mascheroni constant
- $SR_{\mathbf{0}}$ is the Sharpe ratio the luckiest of N worthless strategies is expected to show; it rises with N and with the sampling noise $\sqrt{\mathrm{Var}[\widehat{SR}]}$

The Best of N Zero-Skill Strategies



- Strategies with i.i.d. zero-mean returns, 5 years of daily data (annualised $\mathrm{Var}[\widehat{SR}] \approx 1/5$)
- Trying $N = 1000$ variants yields a best Sharpe ratio of ≈ 1.45 — **with no skill at all**
- Horizontal axis: number of strategies N (log scale); vertical axis: best annualised SR; red line: the theorem; dots: Monte Carlo mean $\pm$ one SD, in line with the theorem

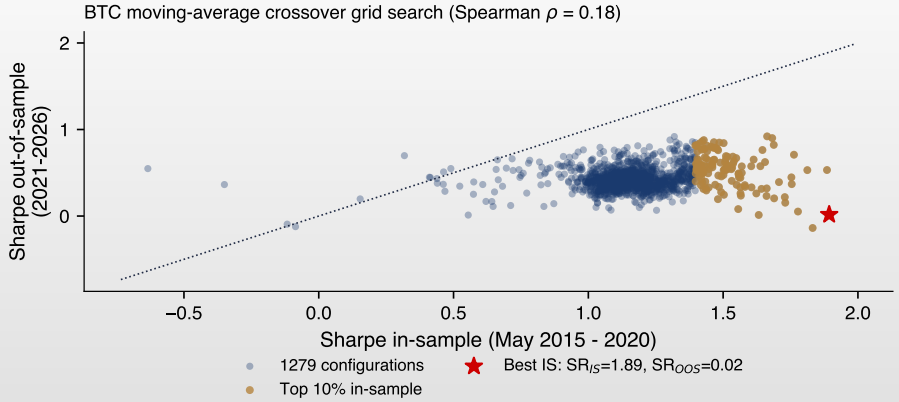
MFM_ch13_backtest_overfitting

In-Sample (IS) vs. Out-of-Sample (OOS): Bitcoin Grid Search (1/2)

- Moving-average (MA) crossover on Bitcoin: long when the fast MA of the log price exceeds the slow MA by more than a band
- 1,279 configurations (fast, slow, band); in sample (IS) 15 May 2015 – 2020, out of sample (OOS) 2021–2026
- Next chart: IS Sharpe ratio on the horizontal axis, OOS Sharpe ratio on the vertical axis; brown points: the top 10% in sample; red star: the selected rule; dotted line: equal SRs
- Selected rule (14/20 days): SR_{IS} **1.89** $\rightarrow$ SR_{OOS} **0.02** (annualised with $\sqrt{365}$)
- Spearman rank correlation between IS and OOS SR: $\rho = 0.18$
- Median OOS SR 0.55 for the top 10% in sample vs. 0.42 for all rules
- The *family* of trend rules carries information; the *selected* optimum is noise

MFM_ch13_backtest_overfitting

In-Sample (IS) vs. Out-of-Sample (OOS): Bitcoin Grid Search (2/2)



MFM_ch13_backtest_overfitting

Sharpe Ratio Inference: Standard Error and PSR (1/3)

- The question: how precise is an estimated Sharpe ratio, and how do we compare it with a benchmark?
- Per-period Sharpe ratio of daily log returns, risk-free rate 0: $\widehat{SR} = \bar{r}/\hat{\sigma}$
- Shape of the distribution, with μ, σ the mean and SD of r: skewness $\gamma_3 = \mathbb{E}[(r-\mu)^3]/\sigma^3$ and ordinary kurtosis $\gamma_4 = \mathbb{E}[(r-\mu)^4]/\sigma^4$ (3 for the Normal distribution)
- Sampling variance of $\widehat{SR}$, i.i.d. returns, delta method (Lo (2002); Opdyke (2007)):

$$\mathrm{Var}(\widehat{SR}) \approx \frac{1}{T}\left(1 - \gamma_3\, SR + \frac{\gamma_4 - 1}{4}\, SR^2\right)$$

  - T: number of returns; SR: true per-period Sharpe ratio
  - negative skewness and fat tails widen the uncertainty; Normal distribution ($\gamma_3 = 0$, $\gamma_4 = 3$): $(1 + \frac{1}{2}SR^2)/T$

Sharpe Ratio Inference: Standard Error and PSR (2/3)

- Annualising with $\sqrt{A}$ needs zero autocorrelation; otherwise $SR_A = \eta(A)\, SR$ (Lo (2002))
  - A: return periods per year (252 trading days; 365 for Bitcoin)
  - $\eta(A) = A/\sqrt{A + 2\sum_{k=1}^{A-1}(A-k)\rho_k}$, ρ_k: autocorrelation of returns at lag k
  - positive autocorrelations, $\rho_k > 0$, make $\eta(A) < \sqrt{A}$
- Example: monthly returns, $A = 12$, with $\rho_1 = 0.2$ and all other autocorrelations zero: $\eta = 12/\sqrt{12 + 2 \cdot 11 \cdot 0.2} = 12/\sqrt{16.4} = 2.96$, against $\sqrt{12} = 3.46$, so the annual SR is about 15% lower
- Two strategies on the same days: test $SR_a - SR_b$ with the HAC delta method or a studentised block bootstrap (Ledoit & Wolf (2008)), not with two separate intervals

Sharpe Ratio Inference: Standard Error and PSR (3/3)

Definition 5 (Probabilistic Sharpe Ratio (PSR), Bailey & López de Prado (2012))

- Called by its authors the probability that the true SR exceeds a benchmark SR^*; formally Φ of a z-statistic, with the variance above in the denominator:

$$\widehat{PSR}(SR^*) = \Phi\left(\frac{(\widehat{SR} - SR^*)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3\,\widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4}\,\widehat{SR}^2}}\right)$$

- SR^*: the benchmark, e.g. 0 or the expected maximum SR_0; $\hat{\gamma}_3$, $\hat{\gamma}_4$: sample skewness and kurtosis
- Reading: $\widehat{PSR} = 1 - p$, with p the one-sided p-value of $H_0 : SR \leq SR^*$
- $\widehat{PSR} > 0.95$ means rejecting H_0 at 5% (i.i.d. returns, delta-method approximation), not a 95% probability for the true SR

Deflated Sharpe Ratio: the Best Bitcoin Rule

- **Deflated Sharpe Ratio** (DSR, Bailey & López de Prado (2014)): $DSR = \widehat{PSR}(SR_0)$, the PSR with the benchmark set to SR_0, the expected maximum of the False Strategy Theorem computed with $\text{Var}[\widehat{SR}_n]$ of the trials
  - corrects for selection (N trials), non-Normality and sample length; reject unless $DSR > 0.95$, and report N
- Worked example, best in-sample rule: $T = 2{,}058$ days, $\widehat{SR} = 0.0991$ a day (1.89 annualised), $\hat{\gamma}_3 = 0.12$, $\hat{\gamma}_4 = 12.33$
  - denominator $\sqrt{1 - 0.12 \cdot 0.0991 + \frac{12.33-1}{4} \cdot 0.0991^2} = 1.008$; $N = 1{,}279$ trials
  - $z = (\widehat{SR} - SR_0)\sqrt{T-1}/1.008$, the argument of Φ; $DSR = \Phi(z)$; $\text{SD} = \sqrt{\text{Var}[\widehat{SR}_n]}$

$\text{Var}[\widehat{SR}_n]$ from	SD	SR_0 (per day)	z / DSR
dispersion of the 1,279 trial SRs	0.0106	0.035	2.88 / 0.998
independent trials, $1/T$	0.0220	0.073	1.16 / 0.877

- Row 1: the observed dispersion of the 1,279 SRs, which includes the correlation between rules
- Row 2: rules treated as independent, Var $= 1/T$, the conservative variant
- The verdict depends on this choice; the out-of-sample SR of 0.02 sides with the conservative value

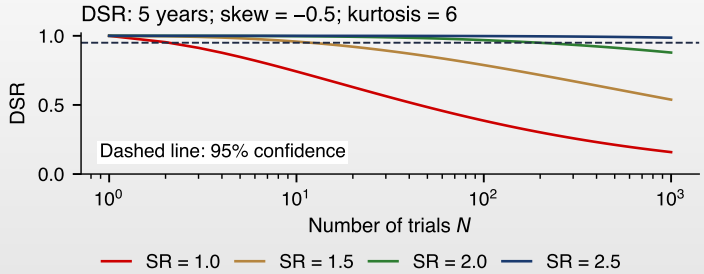
Correlated Trials in the DSR: Alternative Approximations

Two approximations with different assumptions

- (a) raw N, with $\mathrm{Var}[\widehat{SR}_n]$ = dispersion of the trial SRs: row 1 of the previous table
- (b) $\mathrm{Var}[\widehat{SR}_n] = 1/T$, with an effective number of independent trials $\hat{N} < N$: Seminar 13, exercise B6
- Used together, dispersion *and* $\hat{N}$ can count the same dependence effect twice (toy model below)

- Why (a) works, toy model: N zero-skill trials $\widehat{SR}_n = \sigma(\sqrt{\rho}\, Z + \sqrt{1-\rho}\, e_n)$, each of variance $\sigma^2 = 1/T$, independent standard Normal $Z, e_1, \ldots, e_N$, pairwise correlation ρ
  - the common shock moves all trials together, so only $\sqrt{1-\rho}\,\sigma\, e_n$ creates selection: $\mathbb{E}[\max_n \widehat{SR}_n] = \sqrt{1-\rho} \times$ the independent-trial value
  - the dispersion across trials is $(1-\rho)\sigma^2$, so (a) captures this scaling
  - the two rows have SD ratio $0.0106/0.0220 = 0.48$, which this toy model reads as $\rho \approx 1 - 0.48^2 \approx 0.77$
- The dispersion also contains true differences between rules; with real rules (a) is an approximation
- Recommendation: report the conservative row and either (a) or (b), with N; call a rule a discovery only if it also passes the conservative row

Deflated Sharpe Ratio in Practice



- 5 years of daily returns, skewness -0.5, kurtosis 6; horizontal axis: number of trials N (log scale), vertical axis: DSR, dashed line: the 0.95 threshold
- An annualised SR of 1.0 is significant only if (almost) no other strategy was tried: it fails after 2–3 trials
- SR $= 1.5$ survives ≈ 10 trials, SR $= 2.0 \approx 200$; only SR $= 2.5$ survives 1000

MFM_ch13_backtest_overfitting

Data-Snooping Tests on the Bitcoin Grid (1/3) [deeper]

- The question: does any of the K rules beat the benchmark, once we account for having tried them all?
- $r_{k,t}$: return of rule k on day t; $r_{0,t}$: benchmark return; $d_{k,t} = r_{k,t} - r_{0,t}$, with mean $\bar{d}_k$ over the n days
- $H_0 : \max_k \mathbb{E}[d_{k,t}] \leq 0$: no rule beats the benchmark
- Null distribution from a **stationary bootstrap** (Politis & Romano (1994)): resample blocks of whole days, of random length; $\bar{d}_k^*$: the mean in a bootstrap replication
- The same days for all rules keep the correlation between them: no "effective" N needed
- Reality Check (RC, White (2000)): statistic $T^{RC} = \max_k \sqrt{n}\,\bar{d}_k$; its p-value is the share of replications with $\max_k \sqrt{n}\,(\bar{d}_k^* - \bar{d}_k) \geq T^{RC}$

Data-Snooping Tests on the Bitcoin Grid (2/3) [deeper]

- Superior Predictive Ability (SPA, Hansen (2005)): studentised, $T^{SPA} = \max\left(\max_k \sqrt{n}\,\bar{d}_k/\hat{\omega}_k,\ 0\right)$
  - $\hat{\omega}_k$: the bootstrap SD of $\sqrt{n}\,\bar{d}_k$
  - replications are recentred with $\bar{d}_k\,1\{\sqrt{n}\,\bar{d}_k/\hat{\omega}_k \geq -\sqrt{2\log\log n}\}$: clearly bad rules are not recentred, so adding junk rules does not destroy power
- Romano–Wolf StepM (Romano & Wolf (2005)): names *which* rules beat the benchmark, with family-wise error rate (FWER) control, the probability of at least one false rejection
  - step 1: reject $H_{0,k}$ for every k with $\sqrt{n}\,\bar{d}_k/\hat{\omega}_k > \hat{c}_1$, where $\hat{c}_1$ is the $1-\alpha$ quantile of $\max_k \sqrt{n}\,(\bar{d}_k^{*} - \bar{d}_k)/\hat{\omega}_k$
  - step j: recompute the quantile on the rules not yet rejected; stop when no further rule is rejected

Data-Snooping Tests on the Bitcoin Grid (3/3) [deeper]

- Setup: 1,279 rules, in-sample $n = 2{,}058$ days, mean block 20 days, 2,000 replications

Benchmark	largest studentised mean	RC p	SPA p	StepM rules kept
cash ($r_0 = 0$)	3.94	0.004	0.002	314
buy-and-hold	0.88 (the 14/20 rule)	0.74	0.70	0

- Against cash, some trend rule earns a positive mean: Bitcoin's own trend; no rule beats holding Bitcoin, even in sample, consistent with the OOS SR of 0.02
- DSR needs N and $\mathrm{Var}[\widehat{SR}]$, hence its two verdicts; SPA takes both from the data

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Probability of Backtest Overfitting (PBO) [deeper]

Combinatorially symmetric cross-validation (CSCV): Bailey, Borwein, López de Prado & Zhu (2017)

1. Matrix of returns: T periods $\times$ N configurations; split the rows into S contiguous blocks
2. For each of the $\binom{S}{S/2}$ choices, half of the blocks are in sample (IS), the rest out of sample (OOS)
3. Pick the best IS configuration n^*; rank all N configurations OOS in ascending order and set $\bar{\omega} = \text{rank}(n^*)/(N+1) \in (0,1)$
4. Logit $\lambda = \log \frac{\bar{\omega}}{1-\bar{\omega}}$; $PBO = \Pr[\lambda \leq 0]$: the share of splits in which the IS winner ranks at or below the OOS median

- $\bar{\omega}$: relative OOS rank of the IS winner, near 1 if it stays the best; $\lambda > 0$: above the OOS median
- $PBO \approx 0$: the IS winner reliably stays above the OOS median; $PBO \approx 0.5$ or more: selecting the IS winner is no better than picking at random
- PBO evaluates the **research procedure**, not one strategy, and is not the probability of losing money; like DSR it needs **every** trial logged (Seminar 13, exercise B12)

Selection Is Not Inference: Post-Double Selection (1/2)

- The question of this section: how do we get a valid confidence interval for the effect of one variable after ML has chosen the controls?
- The LASSO path answers "which feature enters first", not "which feature matters"; OLS t-statistics after LASSO ignore the selection step
- A control correlated with the regressor of interest d_t but only weakly with y_t gets dropped: its effect moves into the coefficient of d_t (omitted-variable bias)
- Partially linear model: the effect of d_t is linear, the controls enter through unknown functions

$$y_t = \theta\, d_t + g(\mathrm{x}_t) + \varepsilon_t, \qquad d_t = m(\mathrm{x}_t) + v_t$$

  - y_t: outcome; d_t: regressor of interest; θ: its effect, the parameter to estimate
  - x_t: the p controls; g: their effect on y_t; m: how they predict d_t; ε_t, v_t: errors
  - approximately sparse: a few controls explain most of g and m

Selection Is Not Inference: Post-Double Selection (2/2)

- Post-double selection (Belloni, Chernozhukov & Hansen (2014)):
  1. LASSO of y_t on $x_t \Rightarrow$ selected set $\hat{S}_y$
  2. LASSO of d_t on $x_t \Rightarrow$ selected set $\hat{S}_d$
  3. OLS of y_t on d_t and $x_{\hat{S}_y \cup \hat{S}_d}$, robust standard error
- $\hat{S}_y$, $\hat{S}_d$: indices of the controls kept by each LASSO; $x_{\hat{S}_y \cup \hat{S}_d}$: the controls kept by either
- A control is missed only if it matters for *neither* equation, so the omitted bias is second order
- Valid under independent sampling and sparsity, with HAC errors an approximation for time series (penalty and conditions in the Appendix)

Double/Debiased Machine Learning (DML): Orthogonal Scores and Cross-Fitting

- Chernozhukov et al. (2018): residualise both variables with *any* learner (LASSO, RF, boosting), then regress residual on residual

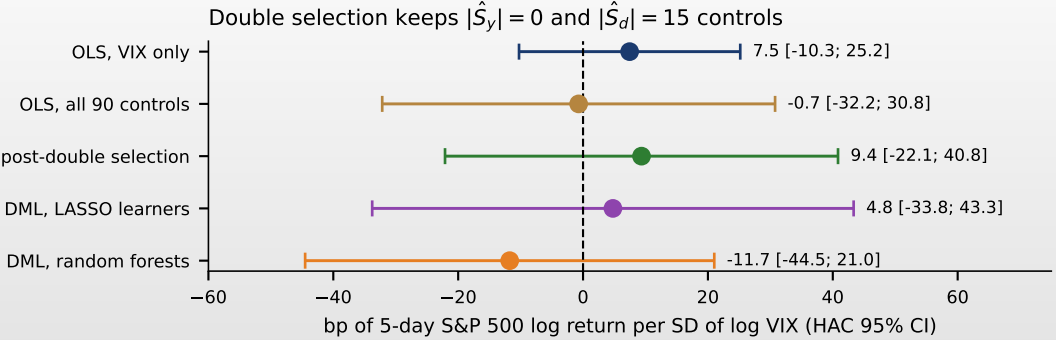
$$\tilde{y}_t = y_t - \hat{\ell}(\mathrm{x}_t), \quad \tilde{v}_t = d_t - \hat{m}(\mathrm{x}_t), \qquad \hat{\theta} = \frac{\sum_t \tilde{v}_t \tilde{y}_t}{\sum_t \tilde{v}_t^2}$$

  - $\hat{\ell}(\mathrm{x}) = \hat{\mathbb{E}}[y \mid \mathrm{x}]$: ML prediction of y_t from the controls; $\hat{m}(\mathrm{x})$: ML prediction of d_t
  - $\tilde{y}_t$, $\tilde{v}_t$: the parts of y_t and d_t the controls do not explain; $\hat{\theta}$: OLS slope of $\tilde{y}_t$ on $\tilde{v}_t$
- Neyman-orthogonal score $\psi = (\tilde{y} - \theta\tilde{v})\,\tilde{v}$, whose mean is zero at the true θ
  - first-order errors in $\hat{\ell}$ and $\hat{m}$ do not bias $\hat{\theta}$ (rate conditions in the Appendix)
- Cross-fitting: fit $\hat{\ell}$, $\hat{m}$ on $K-1$ folds, residualise the held-out fold, repeat; removes the own-observation overfitting bias
- The published theory assumes independent sampling; with overlapping 5-day labels we use contiguous folds purged by 5 days around the test fold and HAC standard errors, an adaptation to serial dependence outside the published theory

Worked Example: Does the VIX Predict S&P 500 Returns? (1/3)

- The question: does the VIX level predict the S&P 500 return over the next 5 days, given a large set of controls?
- y_t: 5-day forward log return, in basis points (bp); d_t: log VIX, standardised (1 SD $= 0.356$)
- 90 controls: 12 technical features, their squares and pairwise products; $n = 6{,}269$ days; HAC errors with 4 lags
- The single LASSO drops the VIX and every control: "no effect", with no standard error
- Double selection keeps $|\hat{S}_y| = 0$ controls in the y_t equation and $|\hat{S}_d| = 15$ in the d_t equation ($|\cdot|$: number of elements)
- Next chart: one estimator per row; the dot is the estimate of θ (bp per week per SD of log VIX), the bar its 95% HAC CI, the dashed line no effect

Worked Example: Does the VIX Predict S&P 500 Returns? (2/3)



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Worked Example: Does the VIX Predict S&P 500 Returns? (3/3)

- Double selection: 95% CI for θ $[-22.1;\ 40.8]$ bp per week per SD of the VIX
- OLS with the VIX only 7.5 $[-10.3;\ 25.2]$; OLS with all 90 controls, double selection and DML give -0.7, 9.4, 4.8 (LASSO) and -11.7 (random forests), all with intervals about twice as wide
- Controls that predict d_t absorb part of the variation of log VIX; less own variation in d_t means a wider interval: the price of protection against omitted-variable bias
- The sign change under DML-RF is noise: every point estimate lies inside the double-selection and DML intervals
- Honest intervals contain zero and large effects of both signs: the data cannot decide

MFM_ch13_inference

The Virtue of Complexity, and a Critical Reading [deeper]

- Kelly, Malamud & Zhou (2024): ridge regressions on P random Fourier features of a few predictors, rolling windows as short as $T = 12$ months, P in the thousands
  - random Fourier features: sin and cos of random linear combinations of the predictors
  - theory: with $P > T$ the minimum-norm solution (the exact fit with the smallest $\|\beta\|_2$) shrinks implicitly ("double descent"); R^2_{OOS} may be negative while the Sharpe ratio of market timing **rises** with complexity; complexity is not automatically overfitting
- Nagel (2025): for $P \gg T$ the forecast is a similarity-weighted average of the T training returns
  - in short windows similarity is mostly recency, so the method becomes a **volatility-timed momentum** strategy, which loses on artificial data with reversals
- Avramov, Cheng & Metzker (2023): deep-learning signal profits come from hard-to-arbitrage stocks
  - excluding microcaps, distressed firms or high-volatility episodes attenuates them, and costs erode them further
- Chen, Pelger & Zhu (2024): impose economic structure instead; the no-arbitrage moment conditions become the loss of a deep network (next slides)
- Survey of the field: Kelly & Xiu (2023); textbook: Dixon, Halperin & Bilokon (2020)

Case Study: Deep Learning in Asset Pricing, Question and Method (1/3) [deeper]

- Chen, Pelger & Zhu (2024): estimate the stochastic discount factor (SDF) M_{t+1} that prices **all individual US stocks**
  - conditioning on 46 firm characteristics and on the history of 178 macroeconomic series
- No arbitrage: the SDF prices every excess return (Section I.A, eq. 1)

$$E_t[M_{t+1}R^e_{t+1,i}] = 0, \qquad M_{t+1} = 1 - \omega_t^\top R^e_{t+1}$$

  - $R^e_{t+1,i}$: excess return of stock i over the Treasury bill from month t to $t+1$; R^e_{t+1}: the vector of all stocks
  - E_t: expectation given the information at t; ω_t: portfolio weights, a function of characteristics and macro states
- $F_{t+1} = \omega_t^\top R^e_{t+1}$, the SDF portfolio, is the conditional tangency portfolio (highest Sharpe ratio), and

$$E_t[R^e_{t+1,i}] = \beta_{t,i}\, E_t[F_{t+1}]$$

  - $\beta_{t,i}$: loading of stock i on F; a stock's expected excess return is its loading times the premium of F (derivation on the next slide)

From the Pricing Condition to the Beta Representation [deeper]

- Three steps (written unconditionally; conditionally, replace E by E_t)
  1. With $M = 1 - F$: $E[(1 - F)R_i^e] = 0$, so $E[R_i^e] = E[R_i^e F]$ for every stock
  2. F is itself an excess return; apply step 1 to $R_i^e = F$: $E[F] = E[F^2]$
  3. Define $\beta_i = E[R_i^e F]/E[F^2]$; then $E[R_i^e] = E[R_i^e F] = \beta_i E[F^2] = \beta_i E[F]$
- This β_i is defined with second moments; when step 1 holds it equals the regression beta, since $\mathrm{Cov}(R_i^e, F) = \beta_i E[F^2] - \beta_i E[F]^2 = \beta_i \mathrm{Var}(F)$; estimated on data with pricing errors, the two can differ, so compute the second-moment form, as the paper's loading network does (▸ Appendix)
- Example: $E[F] = E[F^2] = 0.01$ (1% a month), $E[R_i^e F] = 0.005 \Rightarrow \beta_i = 0.5$, $E[R_i^e] = 0.5\%$ a month; the regression beta gives the same value, $(0.005 - 0.005 \cdot 0.01)/(0.01 - 0.01^2) = 0.5$
- Why the loss changes: any ω that prices all assets must satisfy step 1 for every i, a first-moment condition; this is what the adversarial GMM loss on the next slide enforces

Case Study: Deep Learning in Asset Pricing, Question and Method (2/3) [deeper]

- Why: forecasting (Gu, Kelly & Xiu (2020), the benchmark feedforward network, FFN) fits a second moment where signal-to-noise is low
- The no-arbitrage condition targets a first moment, and averaging over time removes much of the noise
- Method: adversarial generalized method of moments (GMM, eq. 3, after Hansen & Jagannathan (1997))

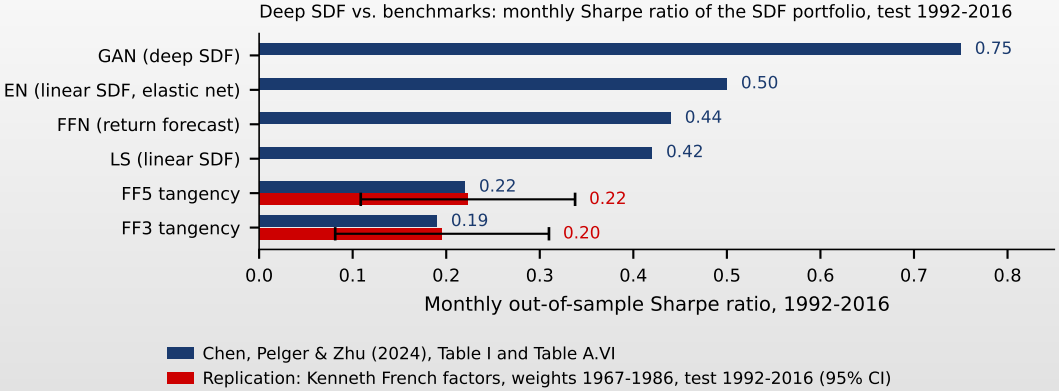
$$\min_{\omega} \max_{g} \frac{1}{N} \sum_{j=1}^{N} \left(E\left[M_{t+1} R^{e}_{t+1,j} \, g(I_t, I_{t,j}) \right] \right)^2$$

  - I_t: macroeconomic information; $I_{t,j}$: characteristics of stock j; N: number of stocks
  - g: instrument function chosen by an adversary to make the pricing errors as large as possible
  - ω minimises this worst case

Case Study: Deep Learning in Asset Pricing, Question and Method (3/3) [deeper]

- Networks
  - an FFN for the weights ω and an adversary g that builds the hardest test assets
  - a long short-term memory network (LSTM) that turns the macro history into 4 states
  - the full model is labelled GAN (Generative Adversarial Network)
- Chronological split: training 1967–1986, validation 1987–1991, test 1992–2016
- Benchmarks: the unregularised (LS) or elastic net (EN) linear SDF and the tangency portfolios of the Fama–French three- and five-factor models (FF3, FF5, Fama & French (2015))
- Configurations ranked by the validation Sharpe ratio (data and tuning in the Appendix)

Deep Learning in Asset Pricing: Out-of-Sample Sharpe Ratios (1/2) [deeper]



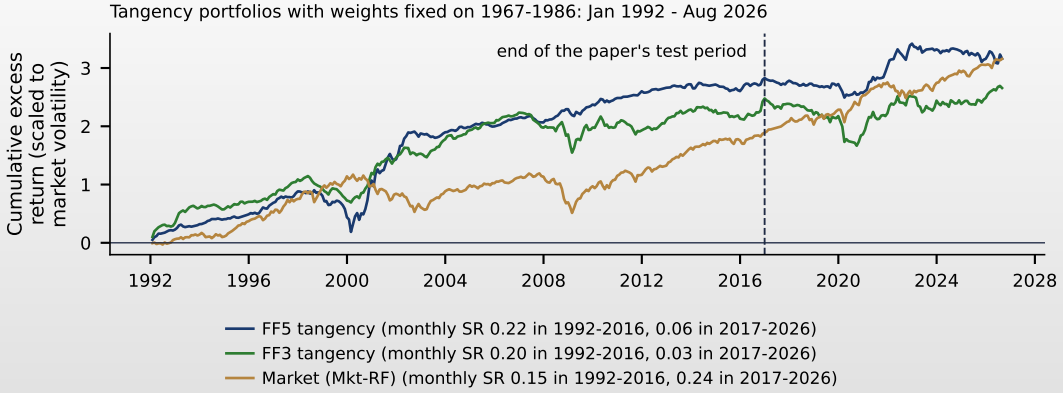
MFM_ch13_sdf_replication

Deep Learning in Asset Pricing: Out-of-Sample Sharpe Ratios (2/2) [deeper]

- Blue: the paper, monthly out-of-sample SR 1992–2016: GAN 0.75, EN 0.50, FFN 0.44, LS 0.42 (Table I)
  - FF5 0.22, FF3 0.19 (Table A.VI)
  - the regularised no-arbitrage model already beats the forecasting network
- Red: FF3 and FF5 tangency portfolios $w \propto \hat{\Sigma}^{-1}\hat{\mu}$ on the Kenneth French factors, weights fixed on 1967–1986: FF5 0.22 [0.11; 0.34], FF3 0.20 [0.08; 0.31] (SE 0.058, Lo (2002))
  - $\hat{\mu}$, $\hat{\Sigma}$: sample mean vector and covariance matrix of the factor returns

MFM_ch13_sdf_replication

Deep Learning in Asset Pricing: the Benchmarks after 2016 (1/2) [deeper]



MFM_ch13_sdf_replication

Deep Learning in Asset Pricing: the Benchmarks after 2016 (2/2) [deeper]

- Same fixed weights, January 1992 – August 2026, each series scaled to the volatility of the market excess return on 1992–2016
  - monthly SR 2017–2026 ($T = 116$, SE ≈ 0.09): FF5 0.06, FF3 0.03, market 0.24 (Mkt-RF in the chart: the market excess return); on 1992–2016: 0.22, 0.20, 0.15
- Replicable with public data: the Fama–French benchmarks
  - not replicable: the stock-level GAN, LS and EN (the panel of the Center for Research in Security Prices, CRSP, is not in the course data)
  - the post-2016 fall is a warning, not yet a test, and the same question applies to a GAN frozen in 1991 (details in the Appendix)

MFM_ch13_sdf_replication

An Open Question: Discovery or Data Mining at Scale?

- Chen, Lopez-Lira & Zimmermann (2022): mining 29,000 accounting ratios for $t > 2$ gives cross-sectional predictability similar to peer-reviewed predictors; for both, about half remains after the original samples
- Novy-Marx & Velikov (2025): an LLM (large language model) pipeline mines over 30,000 signals and keeps 96 that pass a fixed protocol
  - writes hundreds of complete papers, with invented motivations and some imagined citations
  - HARKing (hypothesising after the results are known) at scale
- Open: when a machine proposes features, model and story
  - what is the number of trials?
  - what separates a discovery from the best of many draws?
- The discovery loop: literature → competing hypotheses → data and code → robustness fixed in advance → referee-style critique → report every test (reusable prompts in the Appendix)
- What the human checks
  - every reference resolves (DOI, arXiv)
  - every trial is logged, so M in the correction counts them all
  - correlated tests get a joint null
  - transformations live inside the training fold

Mini-Case: 102 Tests, One Old Survivor (1/2)

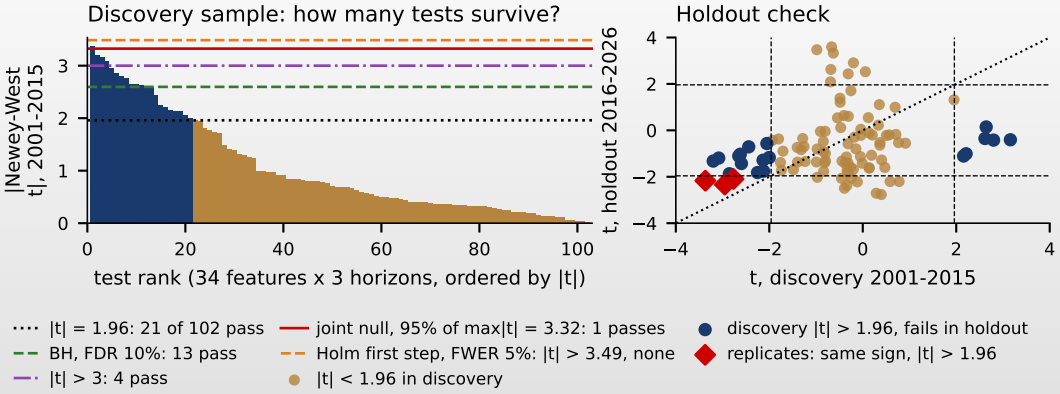
- A typical assistant proposal: "add more indicators: momentum, moving-average distances, volatility ratios, RSI and VIX changes over several windows"
- 34 features $\times$ 3 horizons (1, 5, 20 days) = 102 predictive regressions on 2001–2015, Newey–West t
  - regression j: $r_{t,t+h} = \alpha_j + \beta_j x_{j,t} + e_t$, with $x_{j,t}$ feature j and $r_{t,t+h}$ the next h-day return; $H_{0,j} : \beta_j = 0$
- Corrections for $M = 102$ tests, $p_{(1)} \leq \cdots \leq p_{(M)}$ the ordered p-values
  - Holm: reject while $p_{(j)} \leq 0.05/(M - j + 1)$; controls the FWER
  - Benjamini–Hochberg (BH, Benjamini & Hochberg (1995)): reject the j smallest, for the largest j with $p_{(j)} \leq j\, q/M$
  - controls the false discovery rate (FDR) at q, the expected share of false rejections among all rejections
  - joint null: the 95% quantile of $\max_j |t_j|$ when the target is shifted circularly, which keeps the correlation between the tests

Mini-Case: 102 Tests, One Old Survivor (2/2)

- $|t| > 1.96$: **21** of 102, against 5.1 expected under the global null at 5%
- Holm (FWER 5%): **0**; BH (FDR 10%): **13**; $|t| > 3$ (Harvey, Liu & Zhu (2016)): **4**
- Joint null (500 circular shifts of the target): 95% quantile of $\max_j |t_j| = 3.32$
  - one survivor, the 1-day return on the next day's return, $t = -3.37$: short-term reversal, a known effect
- Holdout 2016–2026: 3 of the 21 keep their sign with $|t| > 1.96$, all reversal variants; the VIX-based "finds" fail
- BH needs independence or positive dependence of the true-null p-values, which correlated predictors do not ensure

MFM_ch13_ai_discovery

Mini-Case: Thresholds and the Holdout (1/2)



MFM_ch13_ai_discovery

Mini-Case: Thresholds and the Holdout (2/2)

- Left: the 102 discovery $|t|$ in decreasing order
- Right: discovery vs. holdout t; the three replications (red) are all short-term reversal; the significant VIX features collapse towards zero

MFM_ch13_ai_discovery

Your Turn: a Project Seed

- **Questions**
  - can an AI-assisted feature search find a predictor of index returns (S&P 500, BET – the Bucharest Stock Exchange benchmark index, Bitcoin)?
  - does it survive a pre-registered correction and a holdout?
  - does it go beyond short-term reversal?
- **Data**: S&P 500 and Bitcoin are in the course data; BET is not, so a team that chooses it names the source (e.g. the Bucharest Stock Exchange), freezes the sample and logs the download date, n and the first and last close
- **Deliverables**
  - a feature list fixed before the data are touched
  - all tests with Holm, BH and the max-$|t|$ null
  - a separate holdout
  - DSR and PBO for any strategy built on the survivors
- **Log** (templates): each prompt that shaped a decision, in `AI_USE.md`
- Each invented reference, leak or undercounted trial in `AI_ERRORS.md`, with the test that caught it

Key Takeaways

Four empirical lessons

- An ADF rejection is not stationarity: truncated FFD of the S&P 500 log price stays $I(1)$; estimate memory directly with local Whittle
- Shuffled K-fold "finds" skill in pure noise
- On the S&P 500 no model beats "always up" (PT, DM, GW)
- The best of many backtests must be deflated or bootstrapped: on the Bitcoin grid the best in-sample rule keeps nothing out of sample

Next

- Chapter 14: deep learning and time-series foundation models
- Chapter 15: large language models and sentiment

A Checklist Before Any Claim

- ☐ Point-in-time data; transformations fitted on training folds only
- ☐ Volatility-scaled labels with recorded spans
- ☐ Purged K-fold or CPCV with embargo for selection; walk-forward for the final test
- ☐ A naive baseline; PT, DM / CW / GW and block-bootstrap intervals
- ☐ Costs included and every trial logged
- ☐ DSR, SPA and PBO for the selected strategy
- ☐ "Selected" or "important" is not "significant": double selection or DML for inference

Check Your Understanding

1. Why does the logit with 57.6% accuracy show no skill when 58.1% of the days are "up"?
2. What does a correlation of 0.98 between the FFD series ($d = 0.25$) and the log price say about an ADF rejection?
3. Why does shuffled K-fold report 69.4% on a target with nothing to predict, while purged K-fold with embargo reports 49.7%?
4. How does SR_0, the expected maximum of N zero-skill strategies, change when N rises from 10 to 1,000?
5. Why does the best Bitcoin rule, with an in-sample SR of 1.89, get a DSR of 0.998 or 0.877 depending on the variance used?
6. Why do the DeLong interval for the AUC, [0.465; 0.494], and the block-bootstrap interval, [0.455; 0.504], lead to different conclusions?
7. Which question about the VIX does double selection answer that the LASSO path does not?

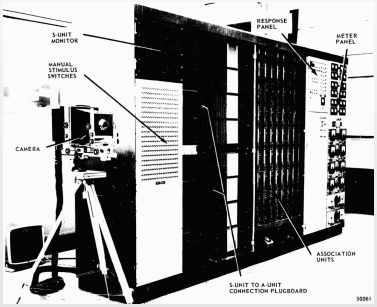
Further Study on Quantinar

Advanced courses on quantinar.com

- Random Forests
- Shapley Values
- Shapley Curves: A Smoothing Perspective
- Machine Learning in Financial Risk
- Introduction to Reinforcement Learning
- Generative Adversarial Networks
- MSCA Digital: The Need for Explainable AI
- Business Analytics and Data Science
- Model Selection Criteria
- PAC Learning

Back

From the Perceptron to Deep Blue



Mark I Perceptron, operator's manual (1960)

Figure: J.C. Hay, A.E. Murray (1960); public domain; Wikimedia Commons



IBM Deep Blue, Computer History Museum

Photo: Anton Chiang (2011); CC BY 2.0; Wikimedia Commons

- **Perceptron** Rosenblatt (1958)
  - a learning rule for a single artificial neuron
  - the Mark I: hardware built to recognise simple images
- 1997: IBM's Deep Blue beats world chess champion Garry Kasparov (Campbell, Hoane & Hsu (2002))
- Deep Blue searched with hand-built rules; modern ML (this chapter) *learns* the rules from data

Back

The In-Sample R^2 of a Predictor without Skill

- A monthly R^2 of 0.5% is economically large, but sampling noise is of the same order
- In-sample OLS R^2 of one predictor with no skill (intercept, independent errors, $n \approx 300$ months): $nR^2 \approx \chi^2_1$
  - n: number of months; χ^2_1: chi-squared distribution with 1 degree of freedom, mean 1 and variance 2
  - so R^2 has mean $1/n \approx 0.3\%$ and standard deviation $\sqrt{2}/n \approx 0.5\%$
- R^2_{OOS} can be negative, has no such null distribution and needs its own test (Clark–West)

Back

Worked Example: How a Forest Averages Its Trees

- Random forest of the case study (400 trees, minimum leaf size 100), trained on the first four purged folds
- Prediction for the last observation (11 September 2026): $P(\text{up in 5 days})$ from each tree
  - individual trees range from 0.31 to 0.88 (standard deviation 0.11)
  - 81.5% of the trees vote "up" ($p > 0.5$)
  - forest probability = average of the 400 trees = 0.608
- Single trees disagree strongly; the average is far more stable
- Stability is not accuracy: the forest still does not beat "always up" (case study)

Back

Sampling Financial Data: Bars

Time and tick bars

- A **bar** = one row (open, high, low, close, volume) summarising a sampling interval
- Time bars: every fixed interval (1 minute, 1 day); oversample quiet periods; fat-tailed returns
- Tick bars: one bar every N transactions

Volume, dollar and information bars

- One bar every V shares or every D dollars traded: sampling by **information arrival**
- Returns closer to i.i.d. draws from the Normal distribution: Clark (1973); Ané & Geman (2000)
- Imbalance / runs bars: sample when order flow becomes one-sided

- This chapter uses daily time bars; tick data (crypto exchanges) in Chapters 10 and 16

Back

Fractional Weights and Their Asymptotics

Definition 6 (Fractional difference operator)

- For a real order d:

$$(1-B)^d = \sum_{k=0}^{\infty} \binom{d}{k} (-B)^k, \qquad \tilde{X}_t = \sum_{k=0}^{\infty} w_k X_{t-k}$$

- Weights: $w_0 = 1$, $w_k = -w_{k-1} \frac{d-k+1}{k} = \frac{\Gamma(k-d)}{\Gamma(-d)\,\Gamma(k+1)} \sim \frac{k^{-(1+d)}}{\Gamma(-d)}$ as $k \to \infty$

- $\binom{d}{k} = d(d-1)\cdots(d-k+1)/k!$: the binomial coefficient, defined for any real d; Γ: the gamma function, $\Gamma(k+1) = k!$
- The weights decay like $k^{-(1+d)}$, more slowly for smaller d
- Infinite sum: $\sum_{k=0}^{\infty} w_k = (1-1)^d = 0$ follows from the binomial series at $B = 1$

Back

Local Whittle and Exact Local Whittle (1/2)

- Periodogram: the sample estimate of the spectral density at the Fourier frequency $\lambda_j = 2\pi j/n$

$$I(\lambda_j) = \frac{1}{2\pi n}\Big|\sum_{t=1}^{n} X_t\, e^{-\mathrm{i}\lambda_j t}\Big|^2$$

- Local Whittle (Robinson (1995)): choose δ to fit $f(\lambda) \approx G\lambda^{-2\delta}$ on the first m frequencies

$$\hat{\delta} = \arg\min_{\delta}\ \log\Big(\frac{1}{m}\sum_{j=1}^{m}\lambda_j^{2\delta} I(\lambda_j)\Big) - \frac{2\delta}{m}\sum_{j=1}^{m}\log\lambda_j, \qquad \sqrt{m}\,(\hat{\delta}-\delta) \xrightarrow{d} N\big(0, \tfrac{1}{4}\big)$$

  - n: sample size; m: number of frequencies used; i: the imaginary unit; G has been profiled out
  - $\xrightarrow{d}$: convergence in distribution; hence the standard error $1/(2\sqrt{m})$

Back

Local Whittle and Exact Local Whittle (2/2)

- Exact local Whittle (ELW, Shimotsu & Phillips (2005); unknown mean: Shimotsu (2010)): periodogram of $(1-B)^{\delta}X_t$, valid also for non-stationary $\delta \geq \frac{1}{2}$
- Why the FFD series looks like $\hat{\delta} \approx 0.75$: above frequency $1/K$ the truncated filter acts as $(1-B)^{0.25}$; only $n/(2\pi K) \approx 2$ of the $m = 294$ frequencies lie below $1/K$, where the unit root returns
  - K: the FFD window length (444 lags)
  - the estimate is the apparent memory over the band, not the integration order of the series

MFM_ch13_ffd

Back

Worked Example: Purging with 10 Observations

- Observations $t = 0, \ldots, 9$; label of t ends at $t + 2$ (capped at 9)
- 5 contiguous folds of 2; embargo = 1 observation

Test	Removed	Train
0, 1	2, 3	4–9
2, 3	0, 1, 4, 5	6–9
4, 5	2, 3, 6, 7	0, 1, 8, 9
6, 7	4, 5, 8, 9	0–3
8, 9	6, 7	0–5

- Test $\{4, 5\}$: labels of 2 and 3 end at 4 and 5 $\Rightarrow$ purged; 6 and 7 start on or before day 7, when the last test label ends $\Rightarrow$ purged; 6 is also the embargo, so here the embargo removes nothing extra ($h = 2 \geq e = 1$)
- With overlapping labels, up to 4 of 10 observations leave the training set

Back

When Is K-Fold Valid for Time Series?

- Bergmeir, Hyndman & Koo (2018): for a **purely autoregressive** model $y_t = g(y_{t-1}, \ldots, y_{t-p}) + \varepsilon_t$ with serially uncorrelated errors, standard K-fold CV is valid
  - g: any, possibly non-linear, function of the last p values; ε_t: error
  - the errors in the test fold are uncorrelated with the training errors, so the CV error is unbiased for the OOS error
- It fails when the **errors** are serially correlated
  - overlapping labels: h-day forward returns share $h-1$ daily returns
  - persistent features that are not in the model's lag set (rolling means, levels such as the FFD log-price)
  - under-specified dynamics (too few lags)
- The leakage experiment isolates exactly this: 20-day labels and 20-day rolling features
- Check before choosing: the autocorrelation of the CV residuals; if it is significant, purge and embargo

Back

Walk-Forward and Combinatorial Purged CV

Walk-forward (WF)

- Mimics real-time deployment; no future data in training
- But a **single path**: high variance, easy to overfit by repeated trials

Combinatorial purged CV (CPCV)

- N contiguous groups; test on all $\binom{N}{k}$ combinations of k groups, with purging and embargo
- $\varphi = \frac{k}{N}\binom{N}{k}$: the number of complete backtest paths; e.g. $N = 6$, $k = 2$: 15 splits, 5 paths
- A **distribution** of Sharpe ratios instead of one number

- Each group is tested φ times, so the test predictions form φ full histories

Back

Testing Directional Accuracy: Pesaran–Timmermann

- H_0: the predicted sign x_t is independent of the realised sign y_t (Pesaran & Timmermann (1992)); $\hat{P}$, $\hat{P}^*$, $\hat{p}_y$, $\hat{p}_x$ as on the main slide, n: number of forecasts

$$PT = \frac{\hat{P} - \hat{P}^*}{\sqrt{\widehat{V}(\hat{P}) - \widehat{V}(\hat{P}^*)}} \xrightarrow{d} N(0,1), \qquad \widehat{V}(\hat{P}) = \frac{\hat{P}^*(1-\hat{P}^*)}{n}$$

  - $\widehat{V}(\hat{P}^*) = \big[(2\hat{p}_y - 1)^2 \hat{p}_x(1-\hat{p}_x) + (2\hat{p}_x - 1)^2 \hat{p}_y(1-\hat{p}_y)\big]/n + 4\hat{p}_y\hat{p}_x(1-\hat{p}_y)(1-\hat{p}_x)/n^2$
  - $\widehat{V}(\cdot)$: estimated variance under H_0; $\hat{P}^*$ is itself estimated, hence the second term
- PT conditions on the marginal frequencies automatically: no need to guess the baseline
- Logit of the case study: $\hat{p}_y = 0.581$, $\hat{p}_x = 0.983$; one-sided $p = 0.90$ for $PT = -1.28$

MFM_ch13_inference

Back

ROC AUC and Its Sampling Uncertainty

- Receiver operating characteristic (ROC) curve: true positive rate $TP/(TP+FN)$ against false positive rate $FP/(FP+TN)$ over all thresholds
- Area under the curve (AUC) $= P(\hat{p}_{\mathbf{pos}} > \hat{p}_{\mathbf{neg}}) + \frac{1}{2}P(\hat{p}_{\mathbf{pos}} = \hat{p}_{\mathbf{neg}}) = U/(n_1 n_0)$
  - $\hat{p}_{\mathbf{pos}}$, $\hat{p}_{\mathbf{neg}}$: predicted probabilities of a randomly drawn positive (up) and negative (down) observation
  - n_1, n_0: numbers of positives and negatives; U: Mann–Whitney statistic, the number of (positive, negative) pairs ranked correctly, ties counted $\frac{1}{2}$
- DeLong standard error (DeLong, DeLong & Clarke-Pearson (1988)): from the placement values, the share of the other class that each observation outranks; assumes independent observations
- Logit, pooled out-of-fold AUC $= 0.479$
  - DeLong 95% CI [0.465; 0.494]: "significantly worse than random"
  - block bootstrap 95% CI [0.455; 0.504]: noise around 0.5
- An AUC below 0.5 is anti-ranking only if it survives a test that respects the dependence

MFM_ch13_inference

Back

Giacomini–White, Statistic and Rolling Window

- Giacomini & White (2006): the null concerns the forecasting **method** (model + estimation window + refitting rule), not the population model

$$H_0 : \mathbb{E}\big[d_{t+\tau} \mid \mathcal{F}_t\big] = 0 \;\Rightarrow\; \mathbb{E}\big[\mathrm{h}_t\, d_{t+\tau}\big] = 0, \qquad GW = n\,\bar{\mathrm{Z}}^{\top}\hat{\Omega}^{-1}\bar{\mathrm{Z}} \xrightarrow{d} \chi^2_q$$

- Notation
  - $d_{t+\tau}$: loss differential of the two forecasts made at t for horizon $\tau = 5$ days; $\mathcal{F}_t$: information available at t
  - h_t: test functions known at t (here 1 and the lagged loss differential, $q = 2$); $\mathrm{Z}_t = \mathrm{h}_t d_{t+\tau}$
  - $\bar{\mathrm{Z}}$: mean of Z_t over the n forecasts; $\hat{\Omega}$: HAC covariance matrix of Z_t
- Reading: under H_0 no information at t predicts which method will do better
- The theory requires a bounded (rolling) estimation window, so that estimation noise never vanishes; an expanding window lies outside the theorem
- S&P 500 logit, rolling 8-year window: GW $= 26.5$ ($p < 0.001$): the logit forecasts worse than the constant

MFM_ch13_inference

Back

Feature Importance: SHAP

Definition 7 (Shapley value of feature j (SHAP, Lundberg & Lee (2017)))

- Average marginal contribution of j over all coalitions $S \subseteq \{1, \ldots, p\} \setminus \{j\}$:

$$\phi_j = \sum_S \frac{|S|!\,(p - |S| - 1)!}{p!} \big[v(S \cup \{j\}) - v(S)\big], \qquad \hat{f}(\mathsf{x}) = \phi_{\mathbf{0}} + \sum_j \phi_j$$

- Interventional SHAP: $v(S) = \mathbb{E}[\hat{f}(\mathsf{x}_S, \mathsf{X}_{\bar{S}})]$, averaging $\hat{f}$ with the features outside S drawn from a background sample; $\phi_{\mathbf{0}} = v(\emptyset)$ = average prediction
- Notation: p: number of features; S: a coalition, a subset of the other features, $|S|$ its size, $\bar{S}$ the features outside it; $v(S)$: prediction when only the features in S are known; $\hat{f}(\mathsf{x})$: model prediction for x
- The weight of S is the probability that exactly the features in S come before j in a random ordering of the p features
- Conditional SHAP uses $v(S) = \mathbb{E}[\hat{f}(\mathsf{X}) \mid \mathsf{X}_S = \mathsf{x}_S]$; the two differ when features are dependent

Shapley Values

Back

Reading Feature Importances

- Remedies for correlated features: permute within clusters, or refit without the feature
- Importances are estimates and need their own uncertainty analysis; honest trees give asymptotically Normal forest *predictions* and CIs (Wager & Athey (2018)), not CIs for importances
- **Backtesting is not a research tool**, importance analysis is: López de Prado (2018)

Back

Double Selection and DML, Penalty and Conditions (1/2)

- Partially linear model: $\mathbb{E}[\varepsilon_t \mid d_t, \mathrm{x}_t] = 0$, $\mathbb{E}[v_t \mid \mathrm{x}_t] = 0$, $\mathrm{Var}(v_t) > 0$; g, m approximately sparse
  - the errors are unpredictable from the regressors; d_t has variation of its own beyond the controls
- Post-double selection: the heteroskedasticity-robust standard error is valid under independent sampling, regularity conditions and sparsity $s^2 \log^2(p \vee n)/n \to 0$ (Belloni, Chernozhukov & Hansen (2014))
  - s: number of controls that matter; p: number of controls; $p \vee n = \max(p, n)$
- LASSO objective, with feature-specific penalty loadings

$$\frac{1}{n}\sum_t (y_t - \mathrm{x}_t^\top \beta)^2 + \frac{\lambda}{n}\sum_j \hat{\psi}_j |\beta_j|, \qquad \lambda = 2c\sqrt{n}\,\Phi^{-1}(1 - \gamma/(2p))$$

  - plug-in constants $c = 1.1$, $\gamma = 0.1/\log n$; loadings $\hat{\psi}_j^2 = \frac{1}{n}\sum_t \mathrm{x}_{tj}^2 \hat{e}_t^2$, $\hat{e}_t$ the residuals of the previous step, iterated

Back

Double Selection and DML, Penalty and Conditions (2/2)

- DML (Chernozhukov et al. (2018)): with $\ell(x) = \mathbb{E}[y \mid x]$, it suffices (with consistency and regularity) that

$$\|\hat{m} - m\| \left(\|\hat{\ell} - \ell\| + \|\hat{m} - m\|\right) = o_P(n^{-1/2})$$

  - $\|\hat{m} - m\|$: root mean squared estimation error of the learner for m; same for ℓ
  - $o_P(n^{-1/2})$: tends to zero in probability faster than $1/\sqrt{n}$
  - e.g. both errors $o_P(n^{-1/4})$: each learner may converge more slowly than a parametric model, because only the product of the errors matters

Back

Machine Learning for Inference in Asset Pricing

- Feng, Giglio & Xiu (2020): does a new factor add pricing information given hundreds of existing ones?
  - double selection of the controlling factors from the zoo, then a valid t-test of the new factor's risk price
  - the inferential answer to Harvey, Liu & Zhu (2016), who raise the hurdle to $t > 3$
- Heterogeneous effects: honest causal forests give CIs for effects that vary with covariates (Wager & Athey (2018))
- Three rules
  - prediction and inference are different targets: a good predictor can give a biased $\hat{\theta}$
  - orthogonalise before you test, and cross-fit with folds that respect time
  - report CIs; "selected" or "important" is not "significant"

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Deep Learning for Sequences

Recurrent networks (LSTM; GRU, gated recurrent unit)

- Hidden state carries memory: $\mathrm{h}_t = g(\mathrm{h}_{t-1}, \mathrm{x}_t)$
  - g: a non-linear function with trained weights, x_t: input at t
- Gates control what is kept or forgotten, mitigating vanishing gradients
- Used for volatility, limit-order-book and intraday forecasting

Transformers

- Self-attention: $\mathrm{softmax}\left(QK^\top/\sqrt{d_k}\right)V$
  - query, key, value matrices Q, K, V (linear maps of the inputs)
  - d_k: key dimension
  - softmax turns each row of scores into weights that sum to 1
- Long-range dependence, parallel training
- Backbone of large language models (Chapter 15)

Back

Time-Series Foundation Models

Pre-trained forecasters

- Pre-trained on large collections of series, used **zero-shot** (without re-training): Chronos (Ansari et al., 2024), TimesFM (Das et al., 2024), Moirai (Woo et al., 2024), Lag-Llama (Rasul et al., 2023)
- Useful baselines for volatility and volume forecasting
- For returns they inherit the same signal-to-noise limits
- Pre-training data may overlap your test period: a new form of leakage

- Covered in depth, with the risk-forecast evaluation tools of Chapter 8, in Chapter 14

Back

Reinforcement Learning, Generative Models, Explainable AI, Risk

Reinforcement learning (RL)

- A policy $\pi(a \mid s)$ chooses actions a in states s to maximise cumulative reward
- Optimal execution, market making, deep hedging: Buehler et al. (2019)

Explainable AI

- Required by regulators (EU AI Act, credit scoring, model risk)
- SHAP, LIME (local interpretable model-agnostic explanations), partial dependence

Generative models

- GANs, variational autoencoders (VAEs), diffusion models: synthetic scenarios for stress tests

ML for risk

- Value-at-Risk (VaR): conditional quantiles (quantile forests, quantile-loss boosting)
- Expected Shortfall (ES): average of fitted tail quantiles or a joint VaR–ES loss
- Volatility: a squared-return target (Chapters 5 and 9)

- **Quantinar:** Reinforcement Learning · GANs · Explainable AI · ML in Financial Risk

Back

Deep Learning in Asset Pricing, Data (Section III.A)

- Monthly returns of all CRSP stocks, January 1967 – December 2016; excess over the one-month Treasury-bill rate
- 46 firm characteristics in six groups: past returns, investment, profitability, intangibles, value, trading frictions (Table A.II)
  - Only stock-months with all 46 available: about 10,000 of 31,000 stocks; ranked each month into cross-sectional quantiles
- 178 macroeconomic series: 124 of FRED-MD, the St. Louis Fed's monthly macroeconomic database (McCracken & Ng (2016)), the 46 monthly medians of the characteristics, 8 predictors of Welch & Goyal (2008); transformed to stationarity

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Deep Learning in Asset Pricing, Method (Sections I–II, Appendix C) (1/2)

- Objective (eq. 3) and networks: main slide; the adversary g uses 8 instruments; three steps: unconditional SDF, maximise over g, minimise again (Section II.D)
- Tuning: 384 configurations ranked by the validation Sharpe ratio (SR); chosen: 2 layers of 64 units, dropout, Adam; ensemble of 9 fits (Table A.VIII); loadings: a second FFN fits $E_t[R^e_{t+\mathbf{1},i} F_{t+\mathbf{1}}] \propto \beta_{t,i}$
- The linear LS/EN benchmark: SDF $\omega_{t,i} = \theta^\top I_{t,i}$ on long and short characteristic legs
  - $\omega_{t,i}$: weight of stock i; $I_{t,i}$: its characteristics; θ: one coefficient per characteristic, the same for all stocks and months

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Deep Learning in Asset Pricing, Method (Sections I–II, Appendix C) (2/2)

- Metrics (II.F): the SR of F and two fit measures; $\hat{\epsilon}_{t+1,i}$: pricing error, the part of $R^e_{t+1,i}$ not explained by $\hat{\beta}_{t,i} F_{t+1}$
- Explained variation: one minus the average squared pricing error over the average squared excess return

$$\text{EV} = 1 - \frac{\overline{\hat{\epsilon}^2}}{\overline{(R^e)^2}}$$

  - overline: time average of the monthly cross-sectional means; EV $= 1$: the factor explains all variation
- Cross-sectional R^2: how well the model explains average returns

$$\text{XS-}R^2 = 1 - \frac{\frac{1}{N}\sum_i \frac{T_i}{T}\left(\bar{\hat{\epsilon}}_i\right)^2}{\frac{1}{N}\sum_i \frac{T_i}{T}\left(\bar{R}^e_i\right)^2}$$

  - $\bar{\hat{\epsilon}}_i$, $\bar{R}^e_i$: time averages for stock i; T_i: months of data on stock i, T: months in the sample

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Deep Learning in Asset Pricing, Results (1/2)

- Source: Chen, Pelger & Zhu (2024), *Management Science*
- Out-of-sample SR of the SDF (Table I), annualised: GAN 2.6, EN 1.7, FFN 1.5 (monthly values on the main slide)
- Individual stocks (Table I): EV 8% (GAN) vs. 3–4%; cross-sectional R^2 23% vs. 14–19%
  - Reading: the no-arbitrage objective improves both fits, although it maximises neither
- Deciles sorted on the loadings β (Table II, test period, annualised): top minus bottom 0.39; FF5 alpha 0.39, $t = 9.96$; the GRS test rejects the CAPM (capital asset pricing model), FF3 and FF5 ($p = 0.00$)
  - GRS (Gibbons–Ross–Shanken) test: H_0 all alphas of the decile portfolios are zero
  - Reading: loadings predict returns, linearly ($R^2 = 0.97$ of mean return on β, Figure 8), and the factor models do not price them

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Deep Learning in Asset Pricing, Results (2/2)

- 46 sets of characteristic-sorted deciles (Table III): GAN cross-sectional R^2 above 90% for all; EV short-term reversal 0.70 vs. 0.43 (EN), momentum 0.54 vs. 0.26
  - Reading: gains come from extreme deciles and interactions
- Macro states (Figure 6): the 178 raw series make the out-of-sample SR collapse; GAN loses about 10% without macro states and 20% with unconditional moments only
  - Reading: the dynamics of macro data must be modelled (LSTM), not fed as the last increment
- Largest stocks (Table IV, monthly SR, models fitted on all stocks, evaluated on the subset)
  - about 1,500 stocks GAN 0.41, FFN 0.30, EN 0.15
  - about 550 stocks 0.26, 0.24, 0.23
  - re-estimated on the 1,500: 0.42, 0.28, 0.42
  - Reading: the edge narrows sharply among the largest stocks; EN catches up when re-estimated on them

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Deep Learning in Asset Pricing, Replication and Extension

- With the Kenneth French Data Library and FRED-MD, on the published split
  - Replicable: FF3 and FF5 tangency weights estimated on 1967–1986 (footnote 23), monthly SR on 1992–2016 next to Table A.VI
  - Not replicable with the course data: the published LS and EN (Tables I and III) use rank-weighted legs of 46 characteristics and stock-level betas
  - Analogue on portfolio data, an extension rather than a replication
  - the linear SDF ($\hat{\theta}_{LS}$ in closed form, $\hat{\theta}_{EN}$ with the objective of Section II.F) on halves (deciles 1–5, 6–10) of the 15 characteristics that the library sorts into deciles
  - its levels are not comparable with Tables I and III

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Deep Learning in Asset Pricing, Limits and an Open Extension

- Limitations stated by the authors
  - The positivity of the SDF is not imposed (footnote 9); only stock-months with all characteristics, which mainly drops very small stocks (Section III.A)
  - Frictions (Section III.I, Figure 17): annual SR 1.73 without the smallest 40% of stocks
  - 2.07 without the widest 40% bid–ask spreads
  - 1.87 without the lowest 40% turnover; see also Avramov, Cheng & Metzker (2023)
  - In-sample metrics exceed the out-of-sample ones, a sign of some overfitting (Section III.C); the hidden states are not comparable across fits, so they cannot be averaged (Section II.E)
- Open extension: freeze the 1967–1991 design and evaluate 2017–2026, after the paper's sample
  - does the linear no-arbitrage SDF still beat FF5?
  - do LSTM states of FRED-MD add to it?

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Deep Learning in Asset Pricing, Reading the Two Charts

- GAN 0.75 vs. FF5 0.22
  - with Lo SEs 0.065 and 0.058, the SE of the difference is at most 0.12 whatever their correlation
  - the gap exceeds 4 SEs (i.i.d. Normal returns)
  - a paired test on aligned returns (Ledoit & Wolf (2008)) is sharper
- The fixed FF5 weights load mainly on profitability and investment: RMW 0.42, CMA 0.30 (weights scaled to $\sum |w| = 1$); FF3 puts 0.61 on HML; both combinations were chosen on 1967–1986

MFM_ch13_sdf_replication

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The Discovery Loop, Step by Step (1/2)

- **Literature**: `"Summarise the evidence on how much published return predictability survives out of sample and after publication: DOI or arXiv id, sample, headline decay estimate."`
  - Resolve every identifier; Semantic Scholar, Google Scholar, Elicit for the search
- **Hypotheses**: `"For a feature that predicts S&P 500 returns in 2001-2015, give competing explanations (risk premium, reversal, microstructure, data mining) and what each predicts for 2016-2026 and for other indices."`
- **Data and code**: `"Write Python that builds 34 technical features from the course's daily S&P 500 and VIX closes with information up to day t only, and reports Newey-West t-statistics for 1-, 5- and 20-day returns in 2001-2015."`
  - Any assistant (Claude, ChatGPT, Gemini, GitHub Copilot)

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The Discovery Loop, Step by Step (2/2)

- **Robustness**, fixed in advance: `"Write the pre-analysis plan: the full list of features and horizons, the holdout, the corrections (Holm, BH, max-t) and the result that would count as a failure."`
  - Placebo: shifting the target circularly (250–3,000 days) breaks its alignment with the features but keeps each series' own dependence
  - a valid null for $\max_j |t_j|$ if the target is stationary and the shifts exceed its dependence range
- **Critique**: `"Act as a referee: which of these significant features are one effect counted several times, and which are known anomalies rather than new ones?"`
- **Report**: all M tests, all trials including the discarded ones, DSR and PBO for any strategy built on them

Back

Acronyms used in this material (1/3)

- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AUC**: Area Under the (ROC) Curve
- **BET**: Bucharest Exchange Trading (index)
- **BH**: Benjamini–Hochberg (procedure)
- **CAPM**: Capital Asset Pricing Model
- **CI**: Confidence Interval
- **CMA**: Conservative Minus Aggressive (investment factor)
- **CPCV**: Combinatorial Purged Cross-Validation
- **CRSP**: Center for Research in Security Prices
- **CSCV**: Combinatorially Symmetric Cross-Validation
- **CV**: Cross-Validation
- **CW**: Clark–West (test for nested models)
- **DM**: Diebold–Mariano (test)
- **DML**: Double/Debiased Machine Learning
- **DSR**: Deflated Sharpe Ratio
- **ELW**: Exact Local Whittle (estimator)
- **ES**: Expected Shortfall
- **EU**: European Union
- **EV**: Explained Variation
- **EWMA**: Exponentially Weighted Moving Average
- **FDR**: False Discovery Rate
- **FF3**: Fama–French three-factor model
- **FF5**: Fama–French five-factor model
- **FFD**: Fixed-width window Fractional Differentiation
- **FFN**: FeedForward Network
- **FN**: False Negative

Acronyms used in this material (2/3)

- **FP**: False Positive
- **FRED-MD**: Federal Reserve Economic Data, Monthly Database (McCracken–Ng)
- **FWER**: Family-Wise Error Rate
- **GAN**: Generative Adversarial Network
- **GB**: Gradient Boosting (the seminar's name for GBM)
- **GBM**: Gradient Boosting Machine (here not geometric Brownian motion)
- **GMM**: Generalized Method of Moments
- **GRS**: Gibbons–Ross–Shanken (test)
- **GRU**: Gated Recurrent Unit
- **GW**: Giacomini–White (test)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HARKing**: Hypothesising After the Results are Known
- **HML**: High Minus Low (value factor)
- **IBM**: International Business Machines (Corporation)
- **IS**: In-Sample
- **LASSO**: Least Absolute Shrinkage and Selection Operator
- **LIME**: Local Interpretable Model-agnostic Explanations
- **LLM**: Large Language Model
- **LS**: Linear SDF estimated by unregularised Least Squares
- **LSTM**: Long Short-Term Memory (network)
- **LW**: Local Whittle (estimator)
- **MA**: Moving Average
- **MA20**: 20-day Moving Average
- **MDA**: Mean Decrease Accuracy
- **MDI**: Mean Decrease Impurity
- **ML**: Machine Learning
- **MLP**: MultiLayer Perceptron
- **MSCA**: Marie Skłodowska-Curie Actions

Acronyms used in this material (3/3)

- **OLS**: Ordinary Least Squares
- **OOS**: Out-Of-Sample
- **PAC**: Probably Approximately Correct (learning)
- **PBO**: Probability of Backtest Overfitting
- **PCR**: Principal Component Regression
- **PLS**: Partial Least Squares
- **PSR**: Probabilistic Sharpe Ratio
- **PT**: Pesaran–Timmermann (test of directional accuracy)
- **RC**: Reality Check (White's test for data snooping)
- **ReLU**: Rectified Linear Unit, $\max(0, z)$
- **RF**: Random Forest
- **RL**: Reinforcement Learning
- **RMW**: Robust Minus Weak (profitability factor)
- **ROC**: Receiver Operating Characteristic (curve)
- **RSI**: Relative Strength Index
- **SD**: Standard Deviation
- **SDF**: Stochastic Discount Factor
- **SE**: Standard Error
- **SHAP**: SHapley Additive exPlanations
- **SPA**: Superior Predictive Ability (test)
- **SR**: Sharpe Ratio
- **TN**: True Negative
- **TP**: True Positive
- **US**: United States
- **VAE**: Variational AutoEncoder
- **VaR**: Value-at-Risk
- **VIX**: CBOE Volatility IndeX
- **WF**: Walk-Forward (validation)

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